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(54) **COMPACT WAVEGUIDE ILLUMINATION SYSTEM**

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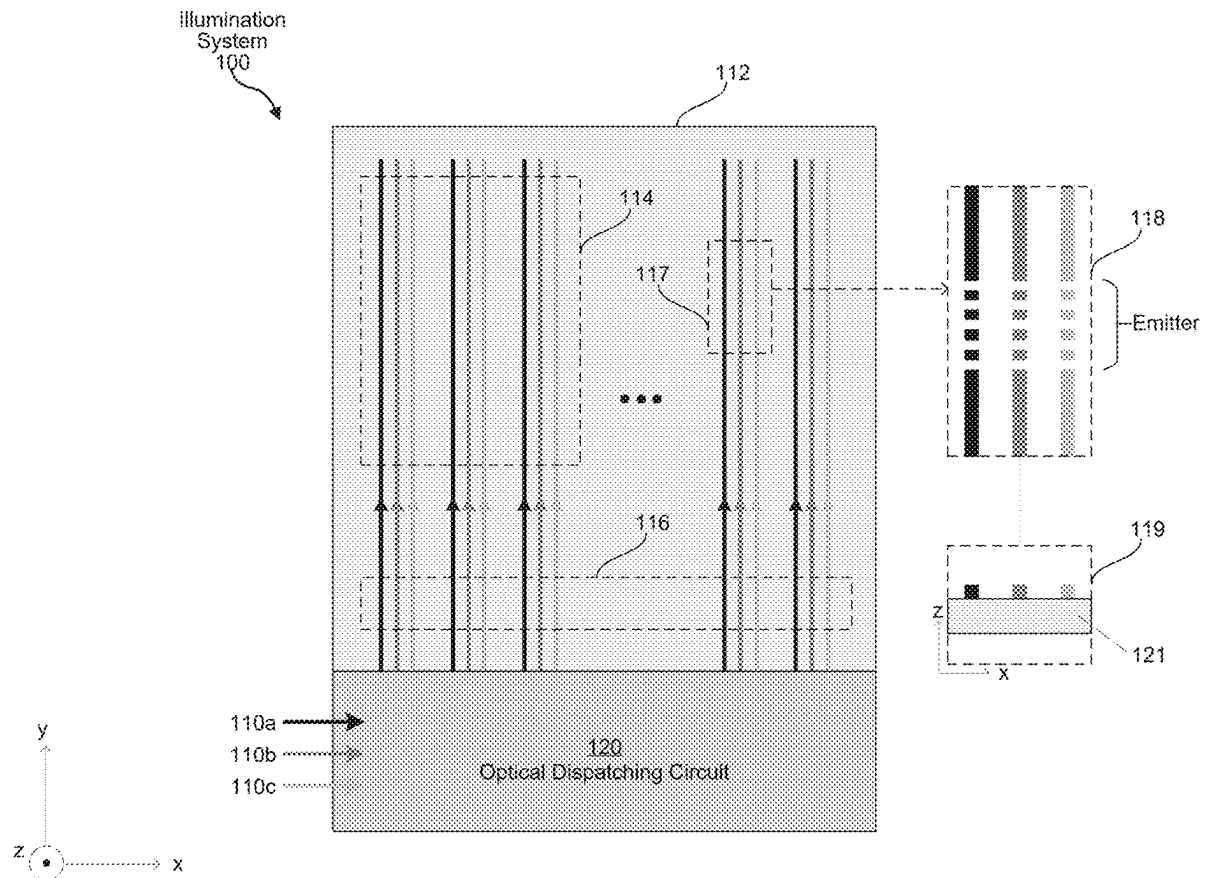
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(2013.01); **G02B 27/0093** (2013.01); **G02B**
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(57)

ABSTRACT

The present application is directed to optical dispatching circuits that may reduce the footprint of an illumination system. In particular, embodiments of the present application provide an illumination system that splits and spreads incoming light sources (e.g., RGB laser light sources) into a plurality of emitters that cover a two-dimensional (2D) area. The present application describes various implementations of an optical dispatching circuit, which receives light as input and is configured to spread this light across a number of waveguides that each emit light from a plurality of locations. The optical dispatching circuits described herein may be configured to receive light from multiple sources emitting at different wavelengths (such as, but not limited to, red, green and blue light) and effectively deliver the light from the multiple sources in a substantially uniform manner to a plurality of emitters that cover a 2D area.



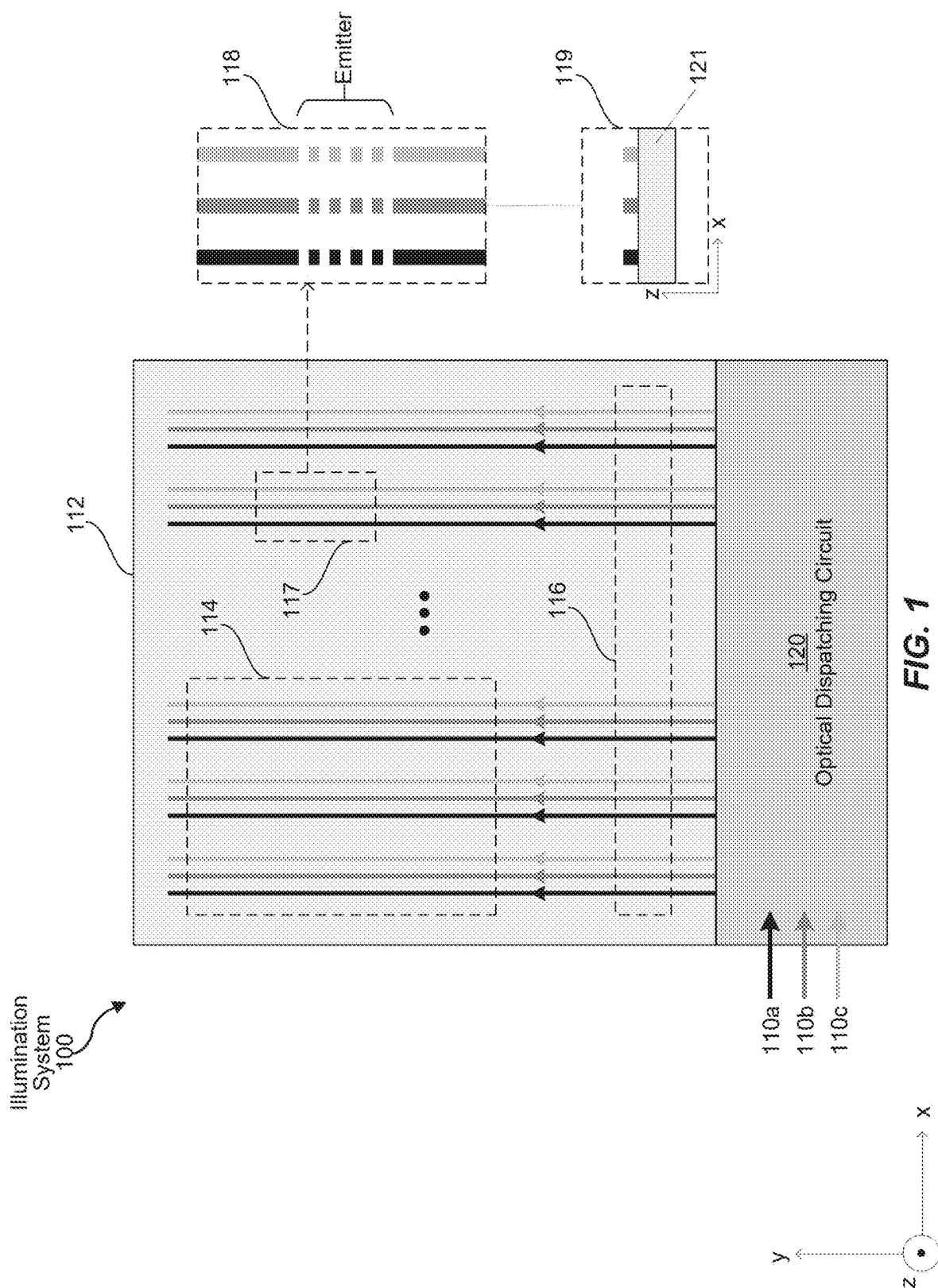


FIG. 1

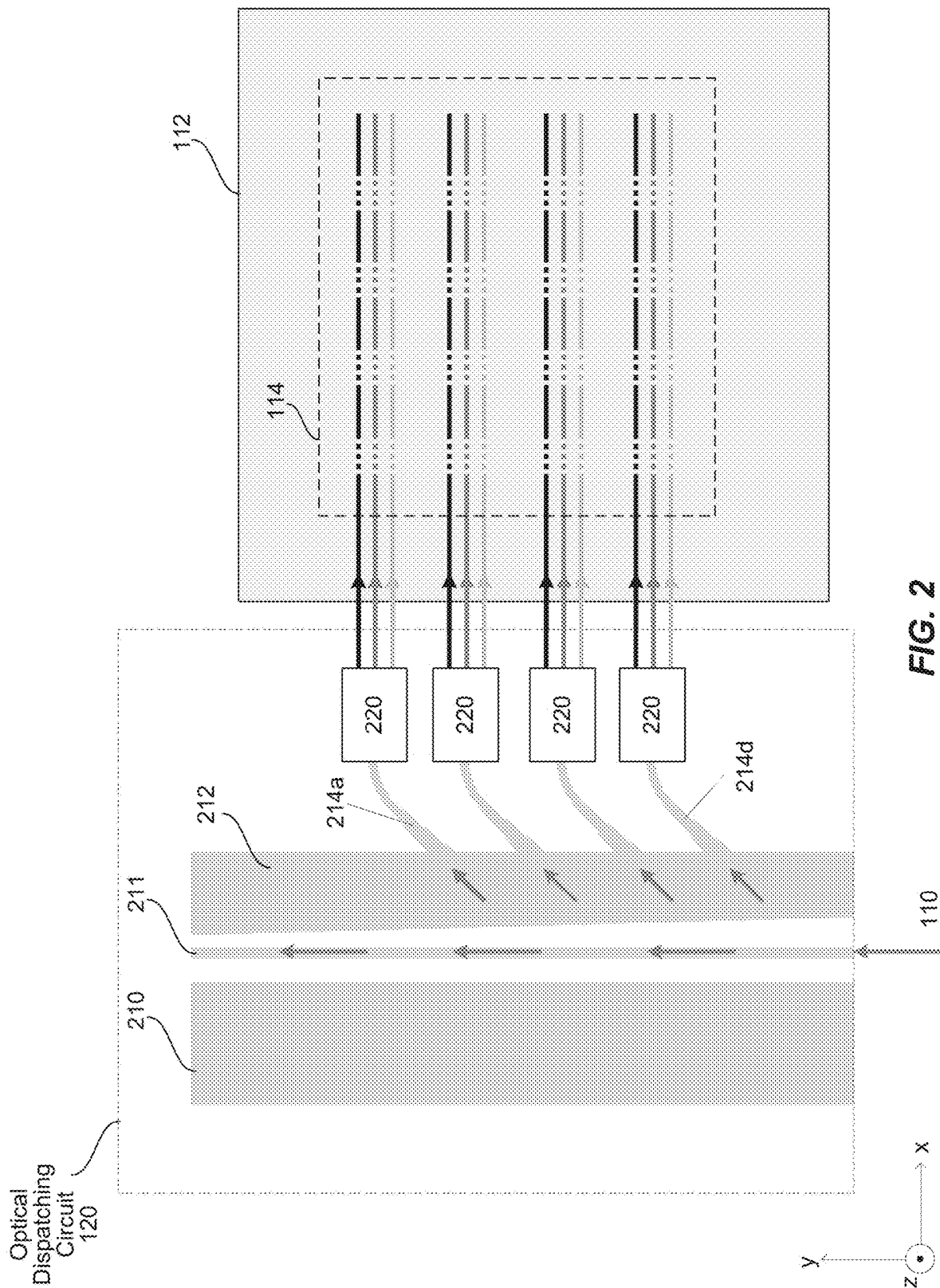


FIG. 2

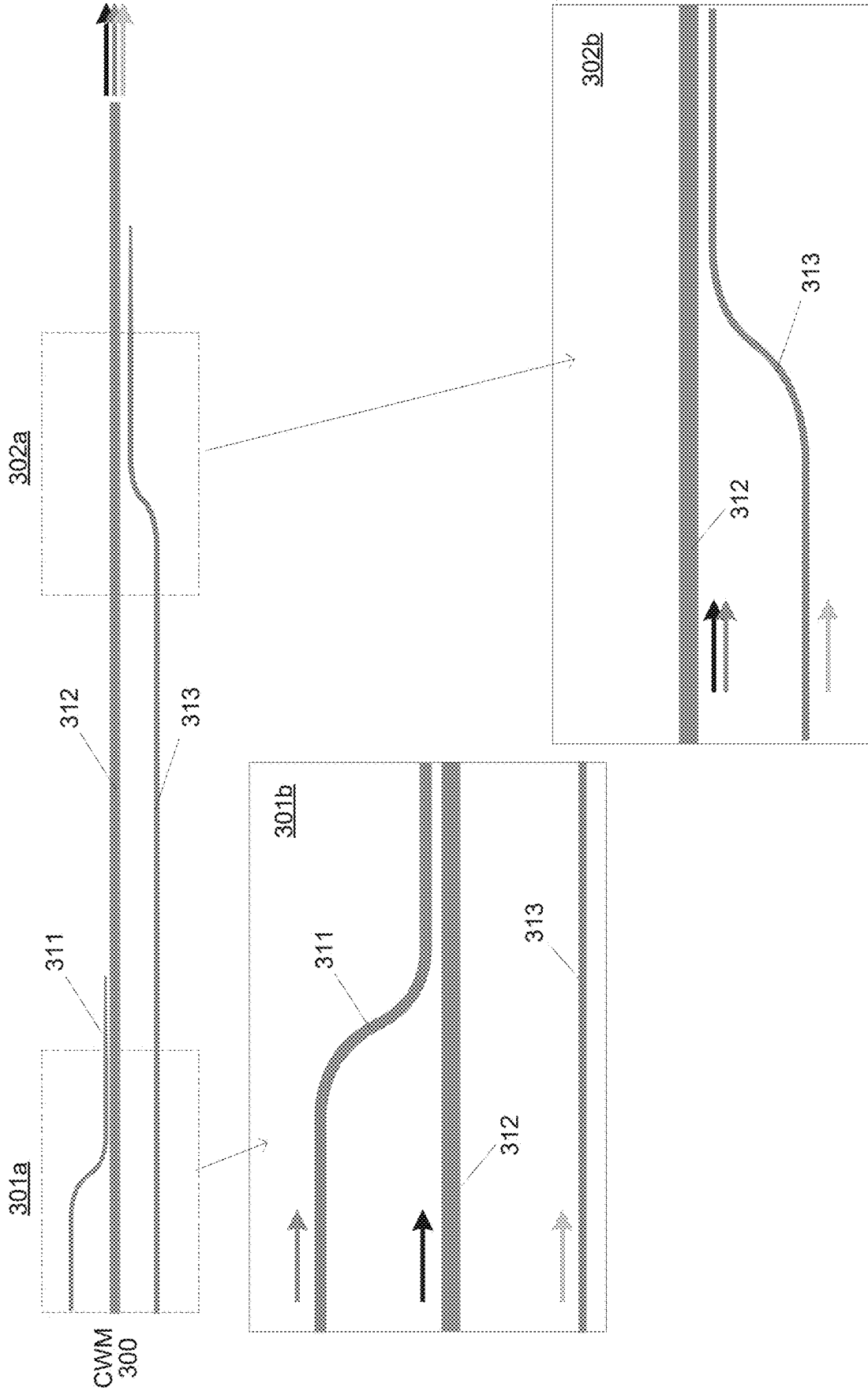


FIG. 3

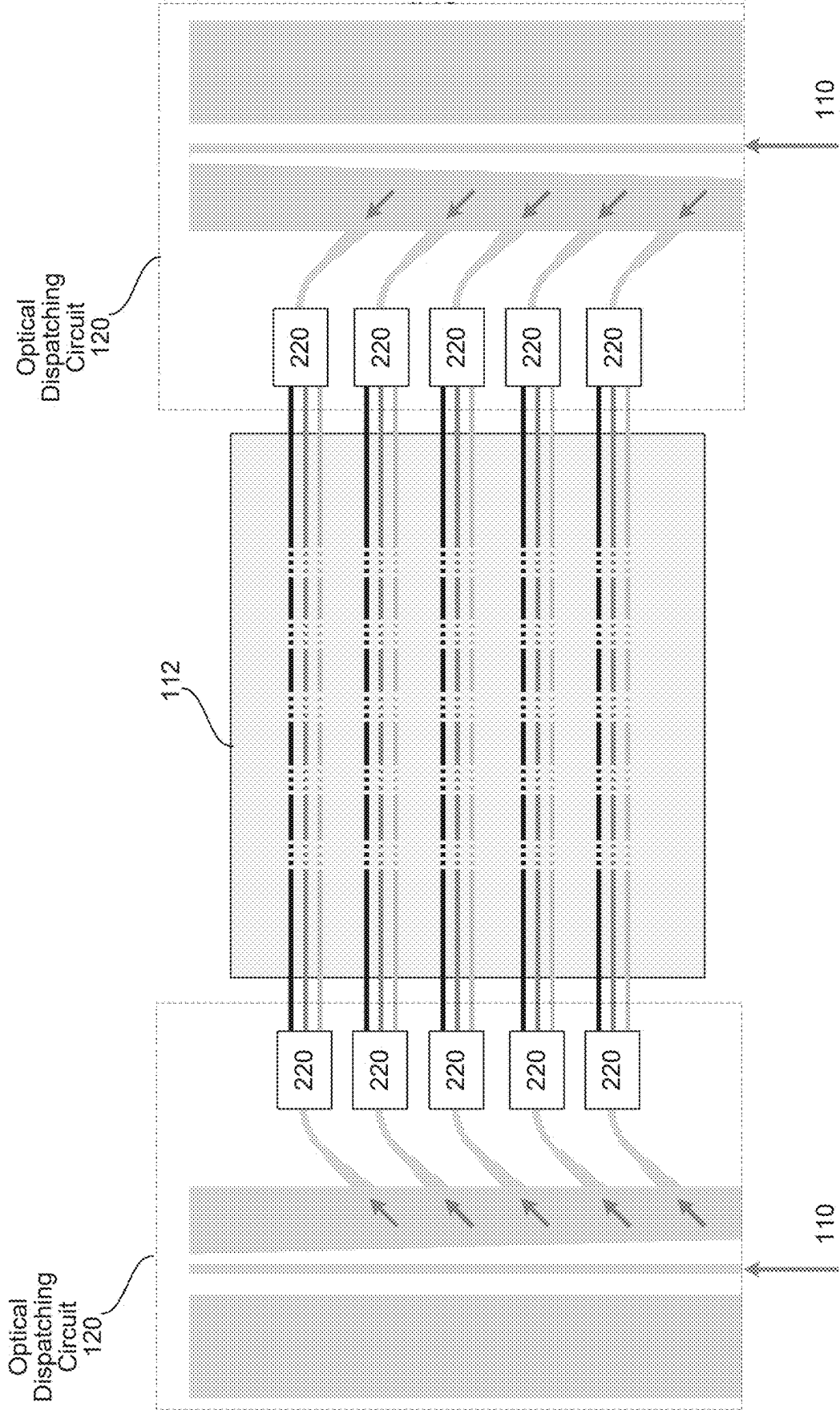
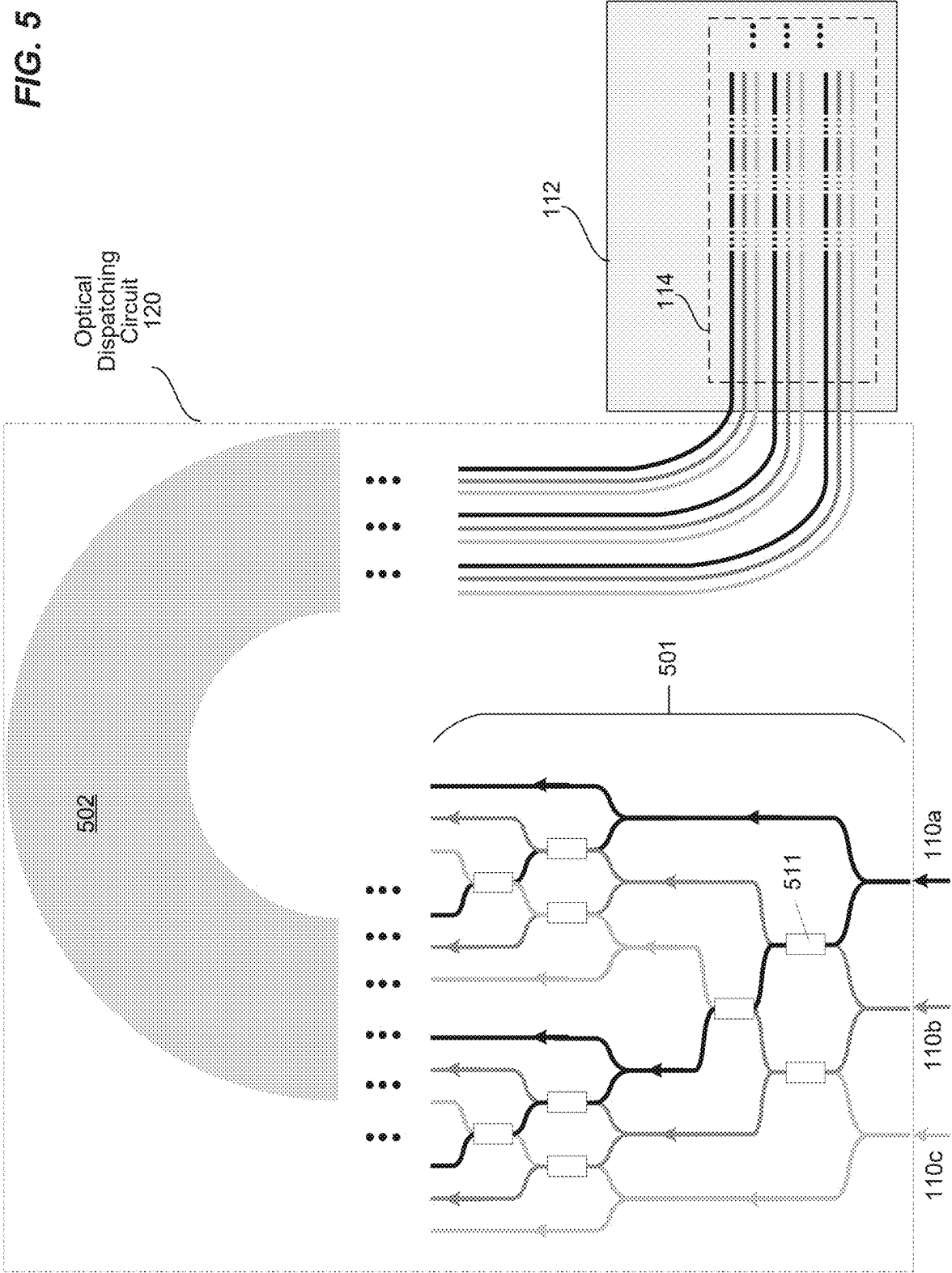


FIG. 4

FIG. 5



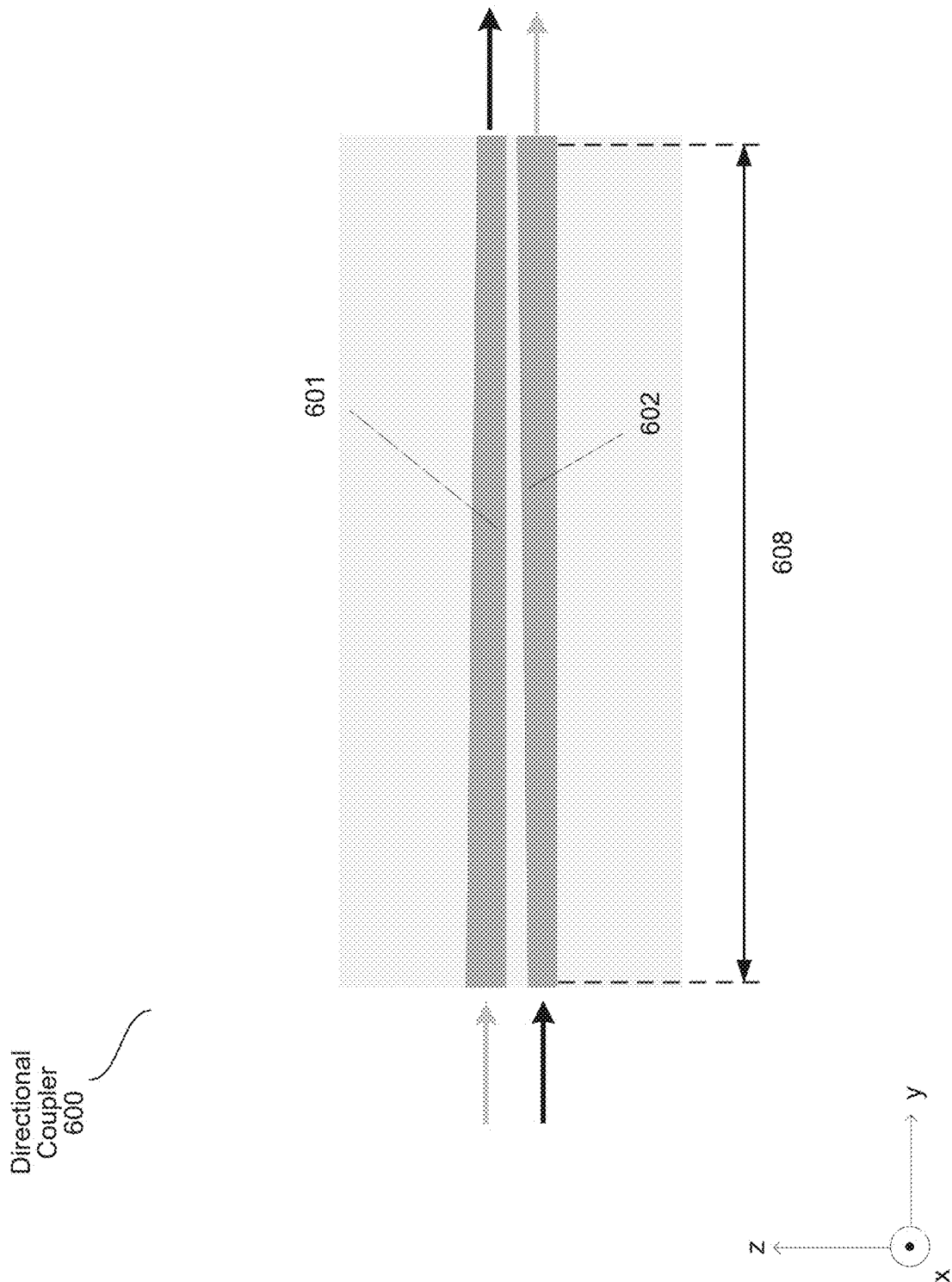


FIG. 6

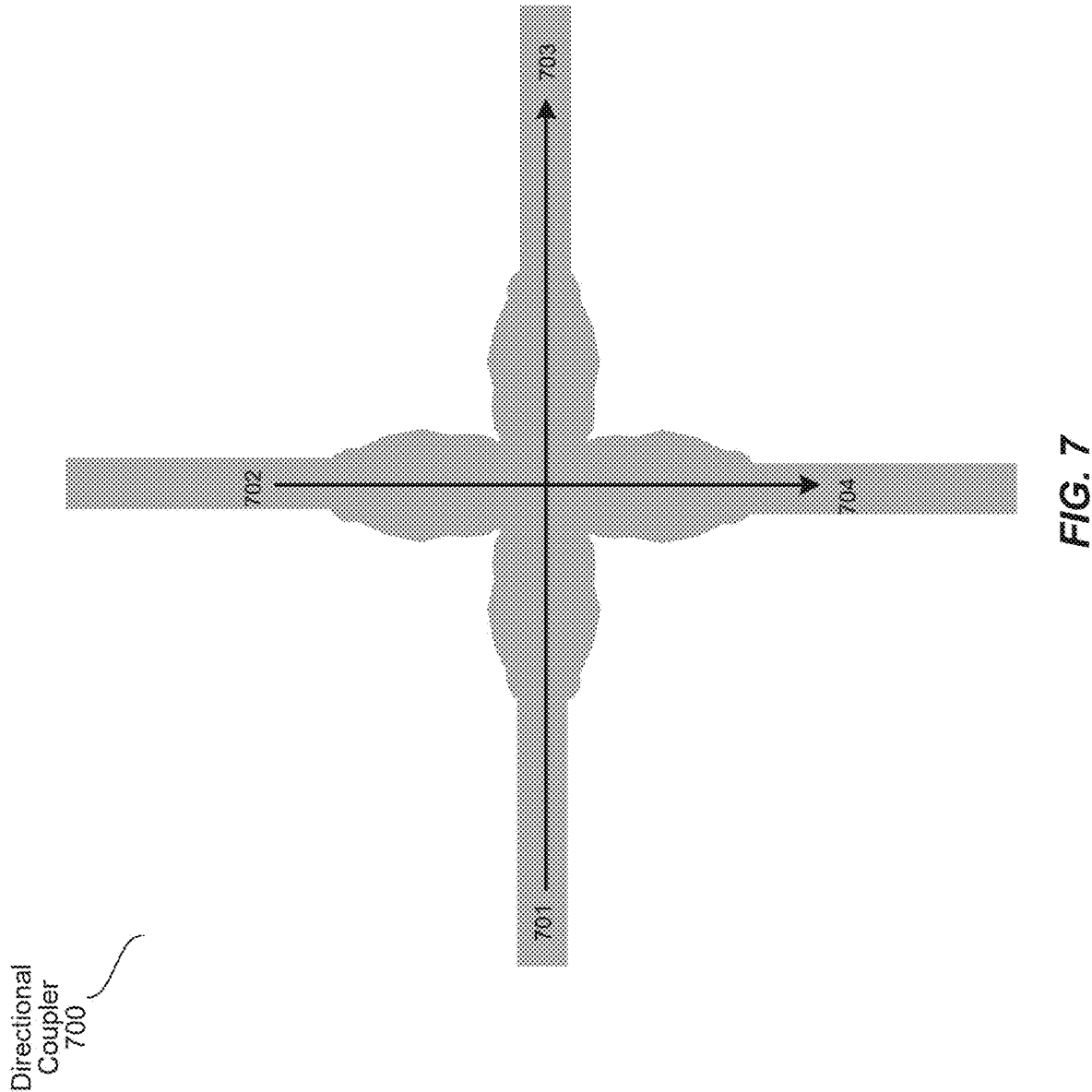
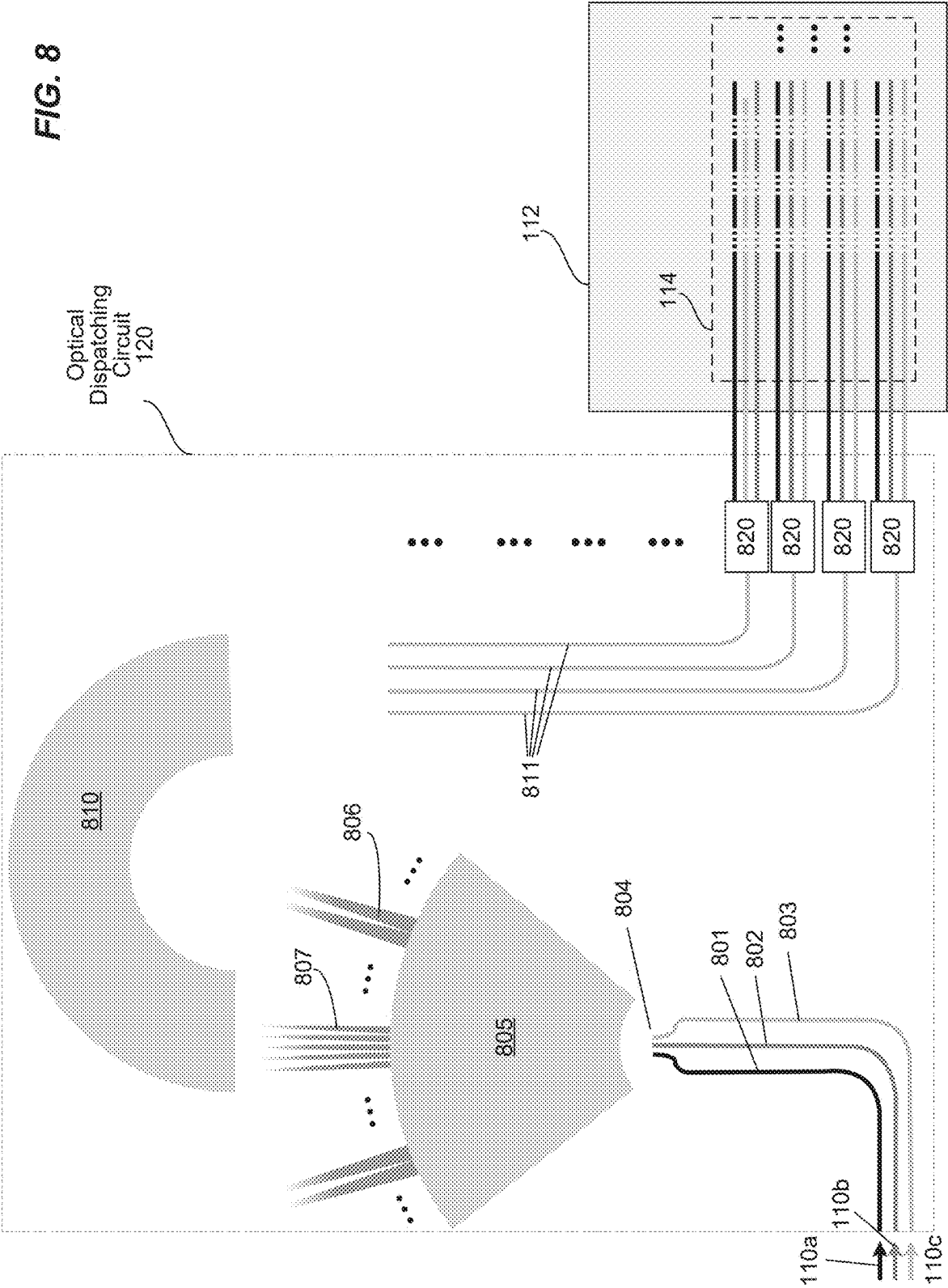


FIG. 8



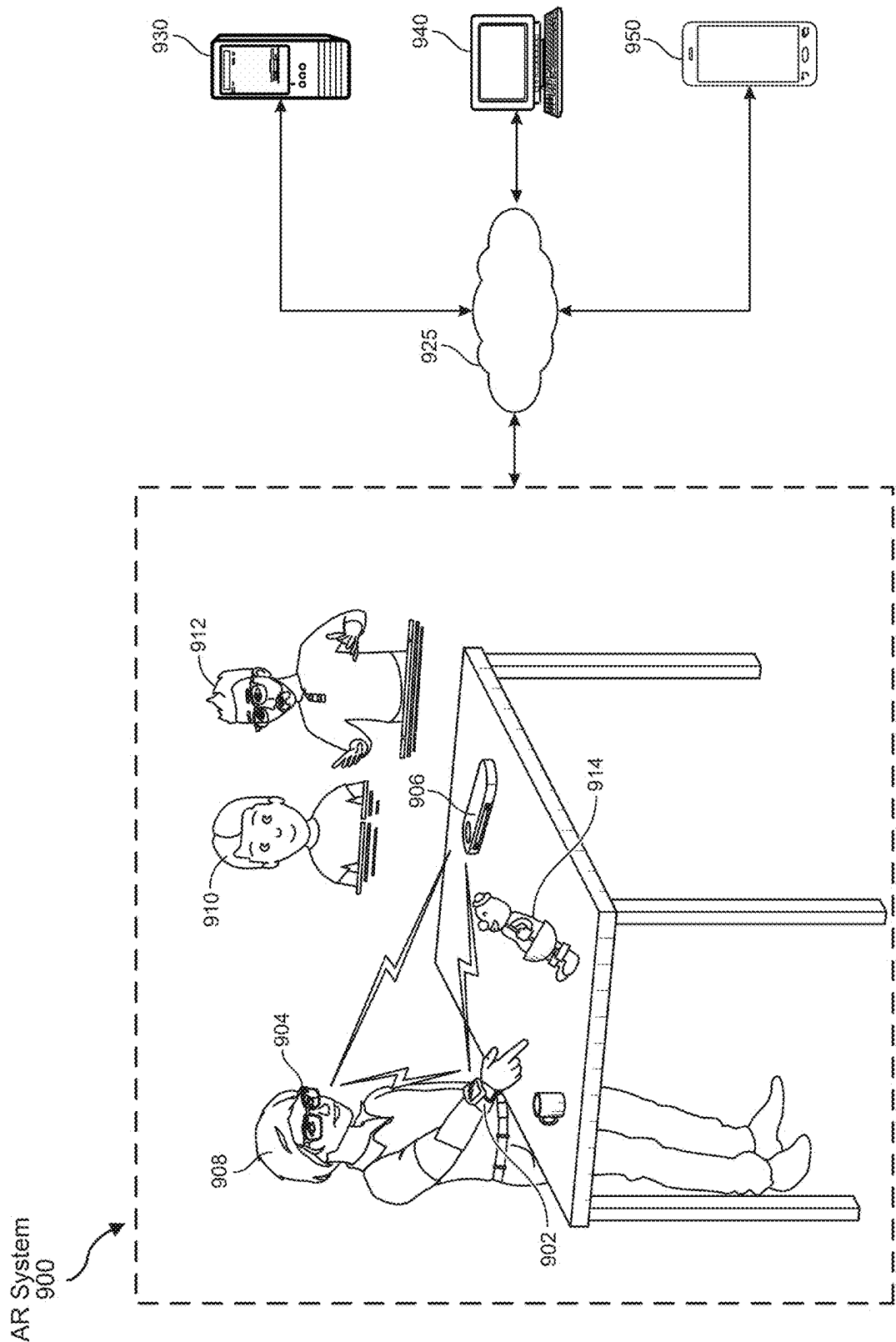


FIG. 9

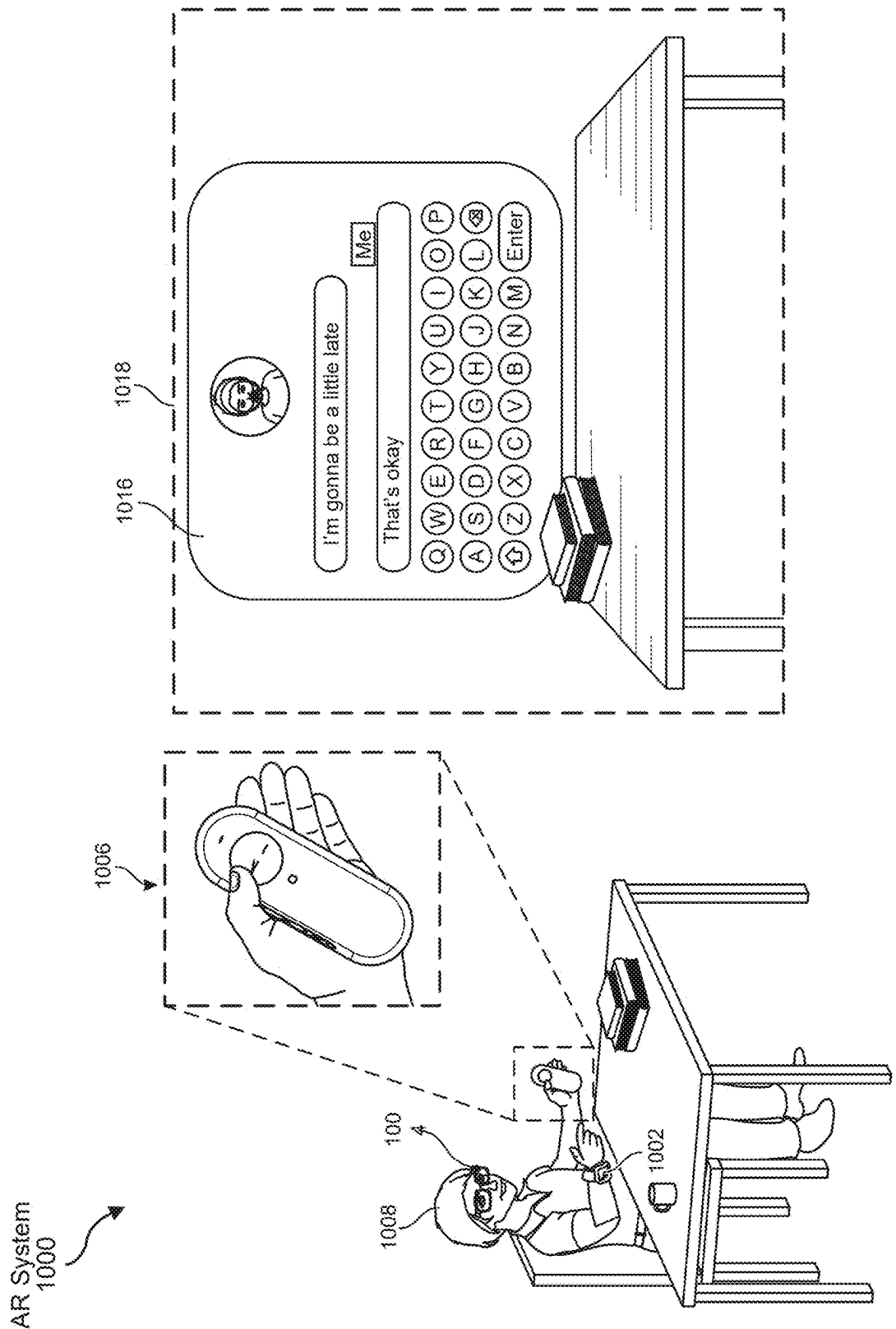


FIG. 10

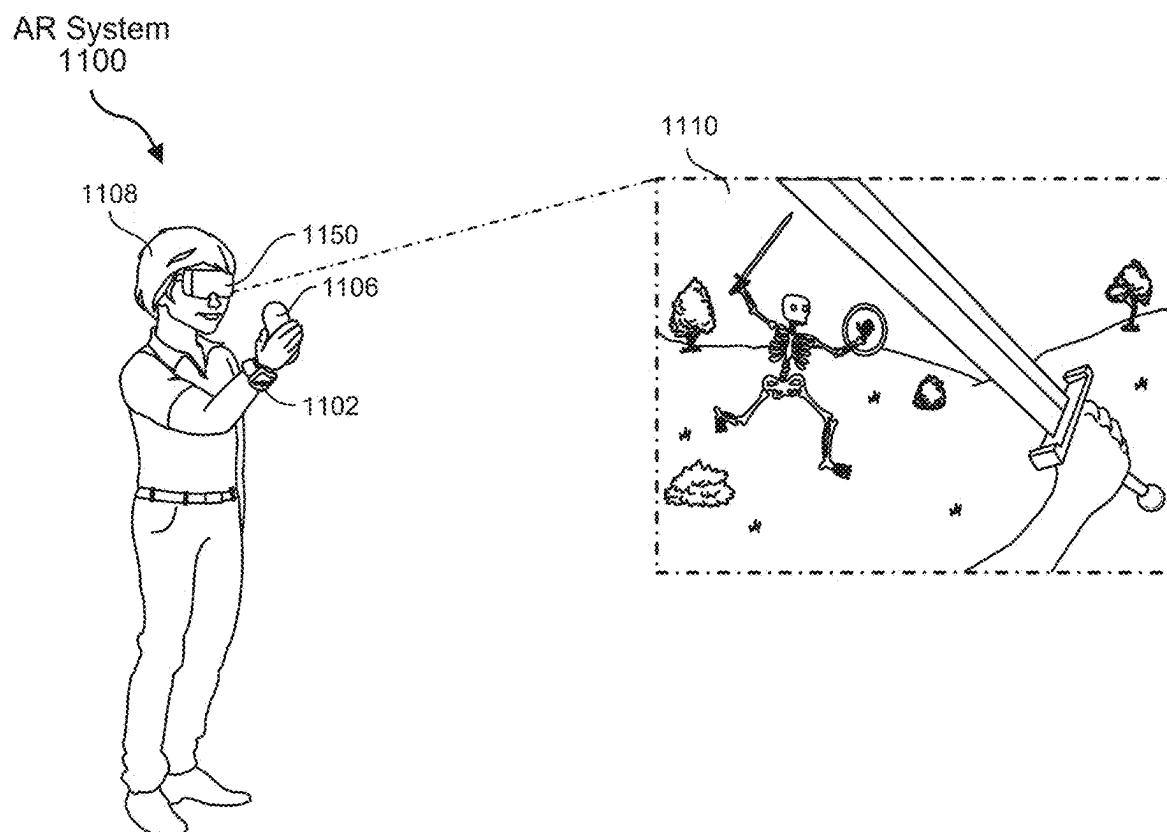


FIG. 11A

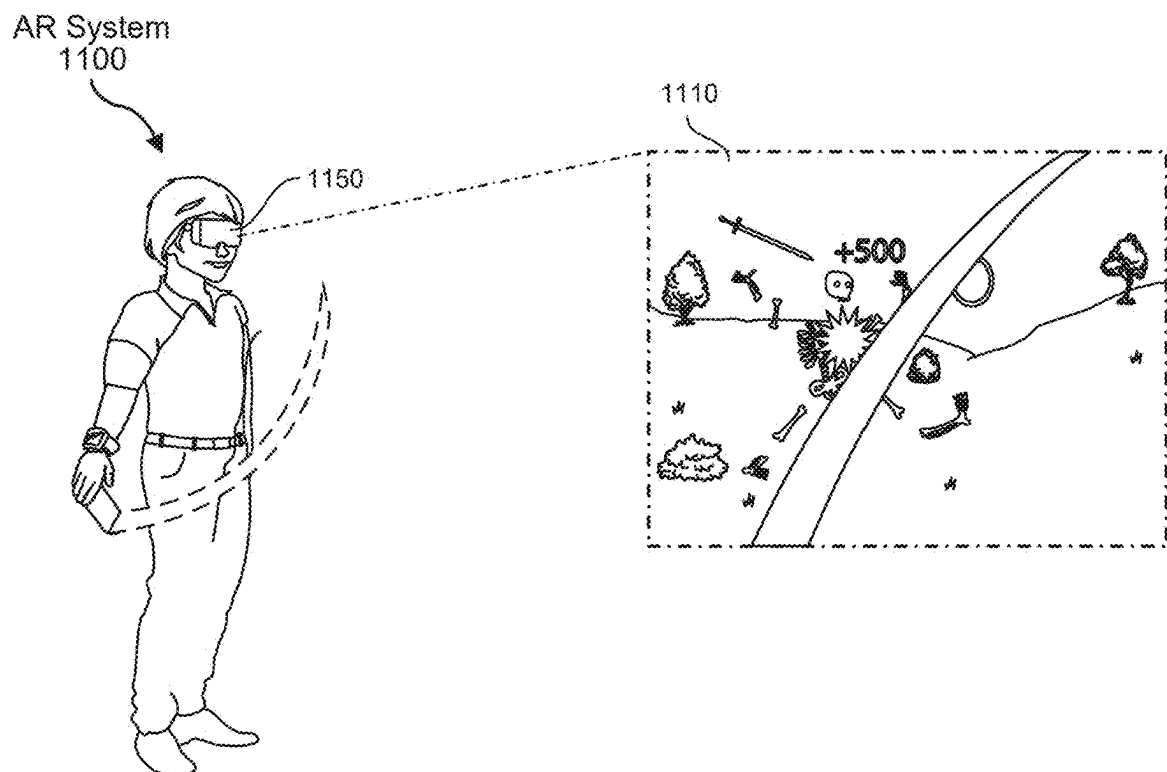


FIG. 11B

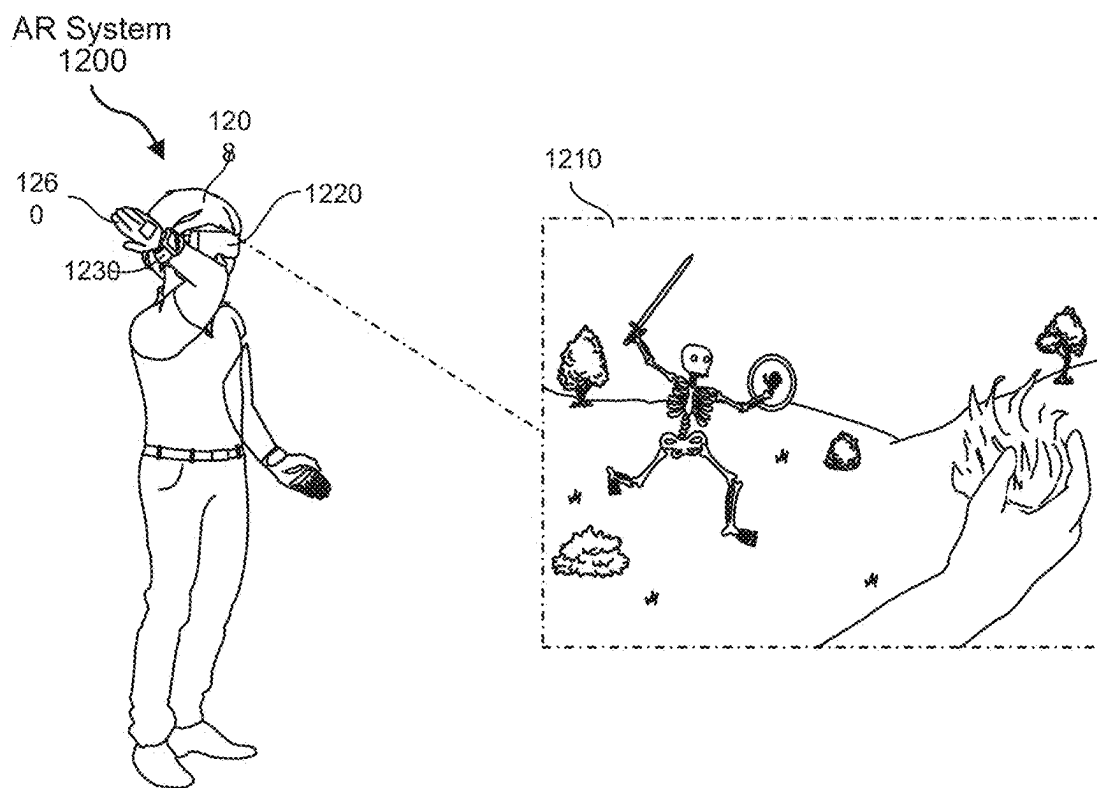


FIG. 12A

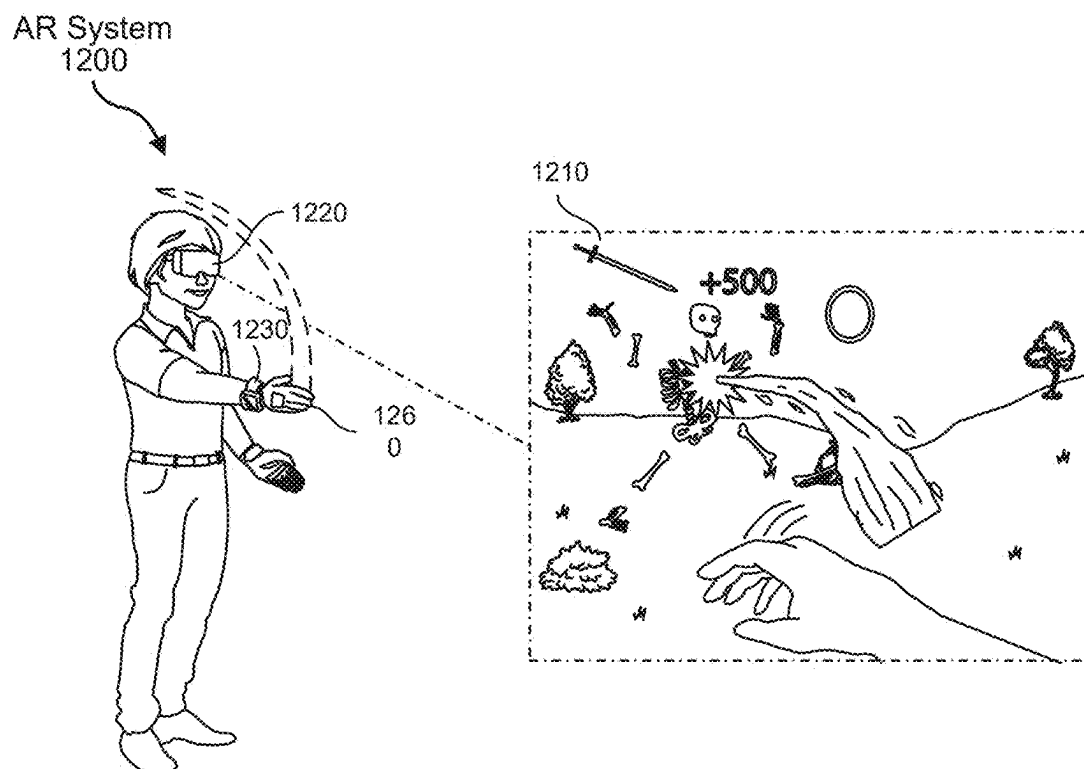


FIG. 12B

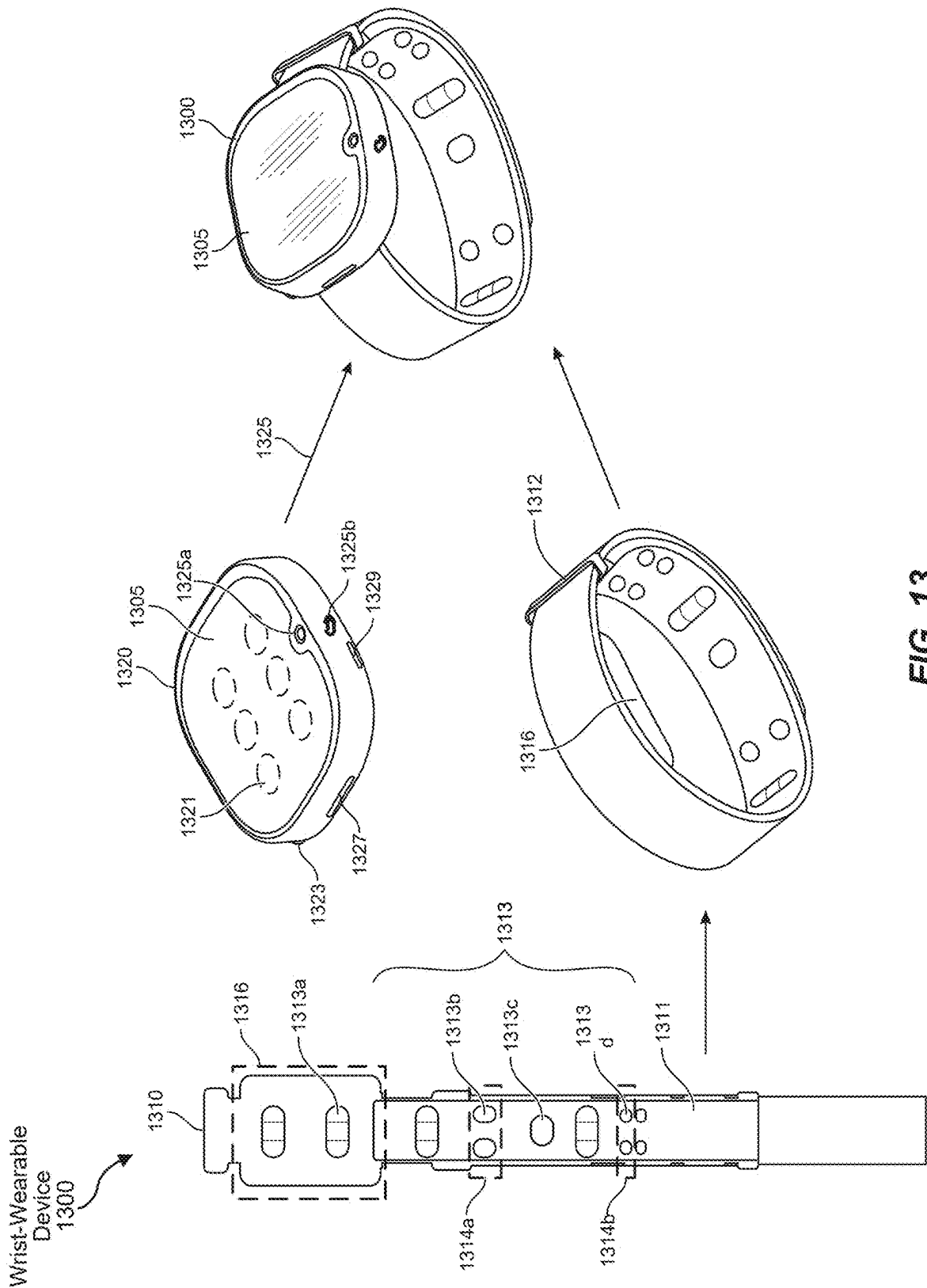


FIG. 13

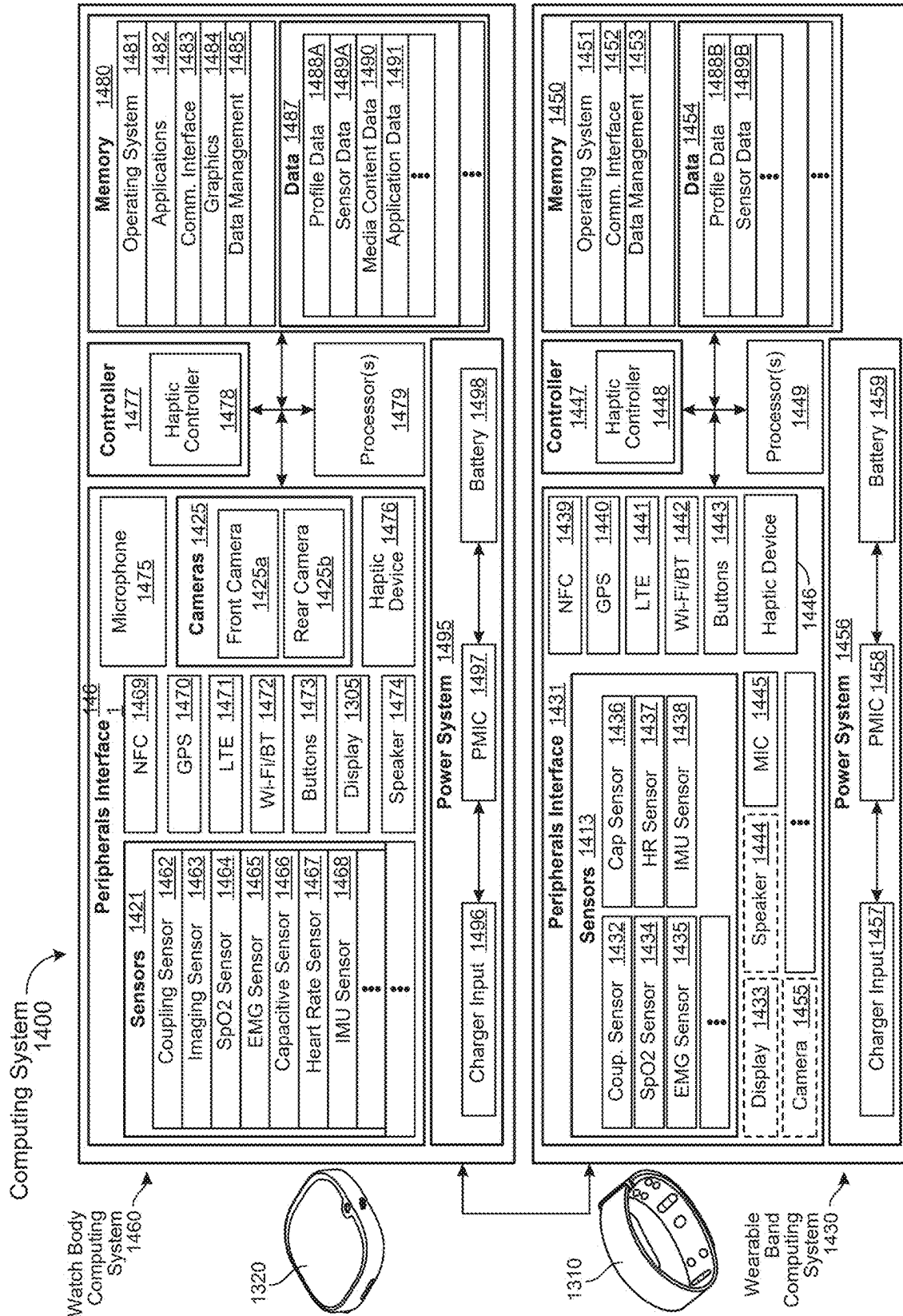


FIG. 14

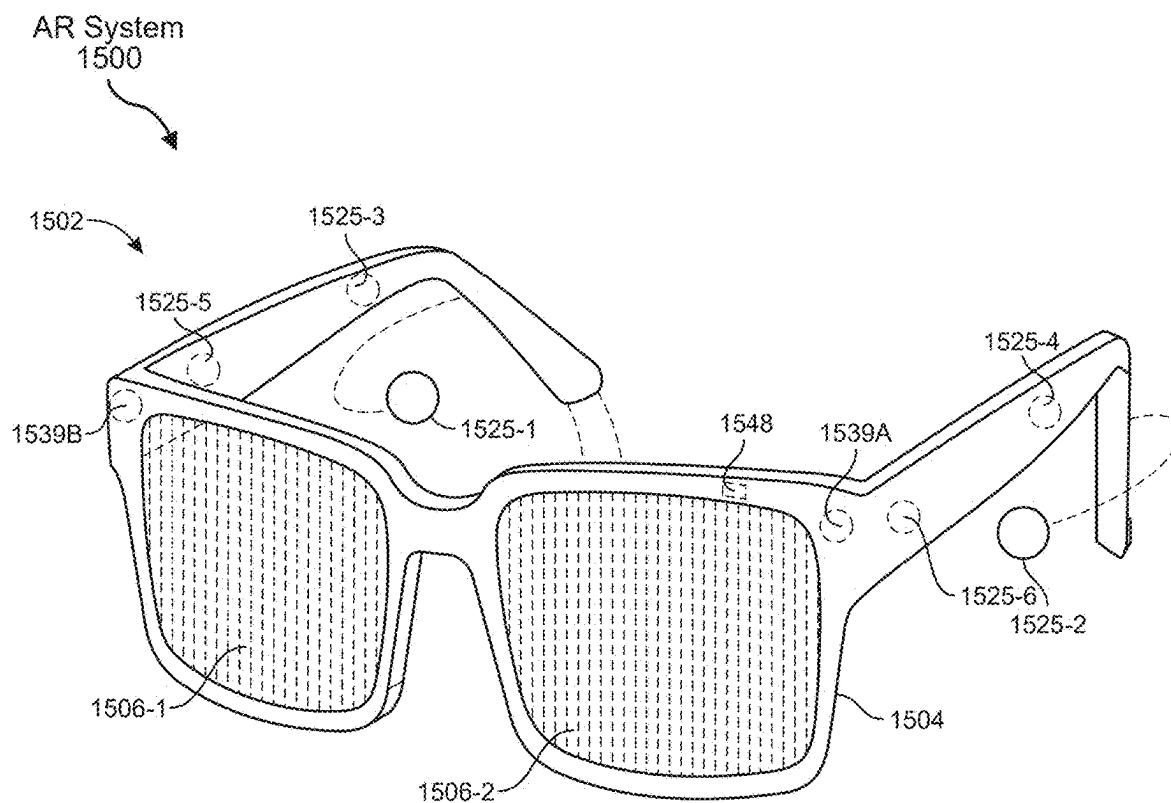


FIG. 15

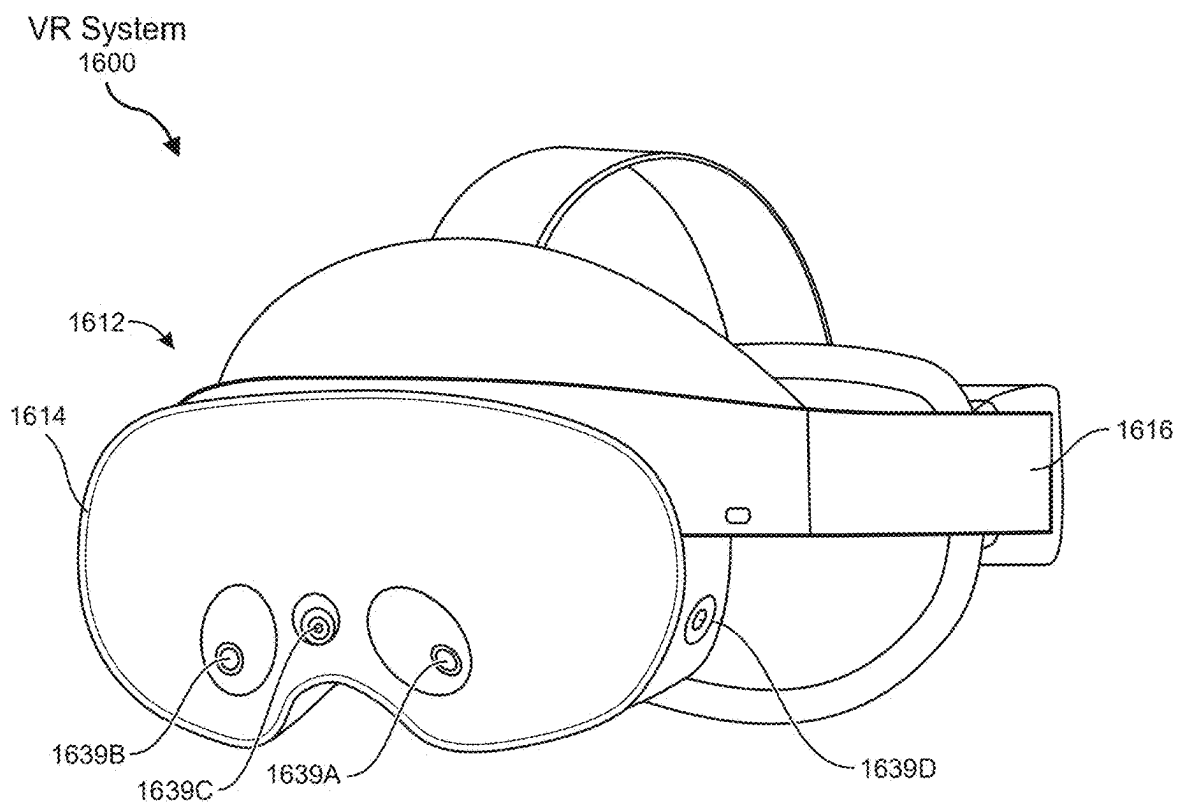


FIG. 16A

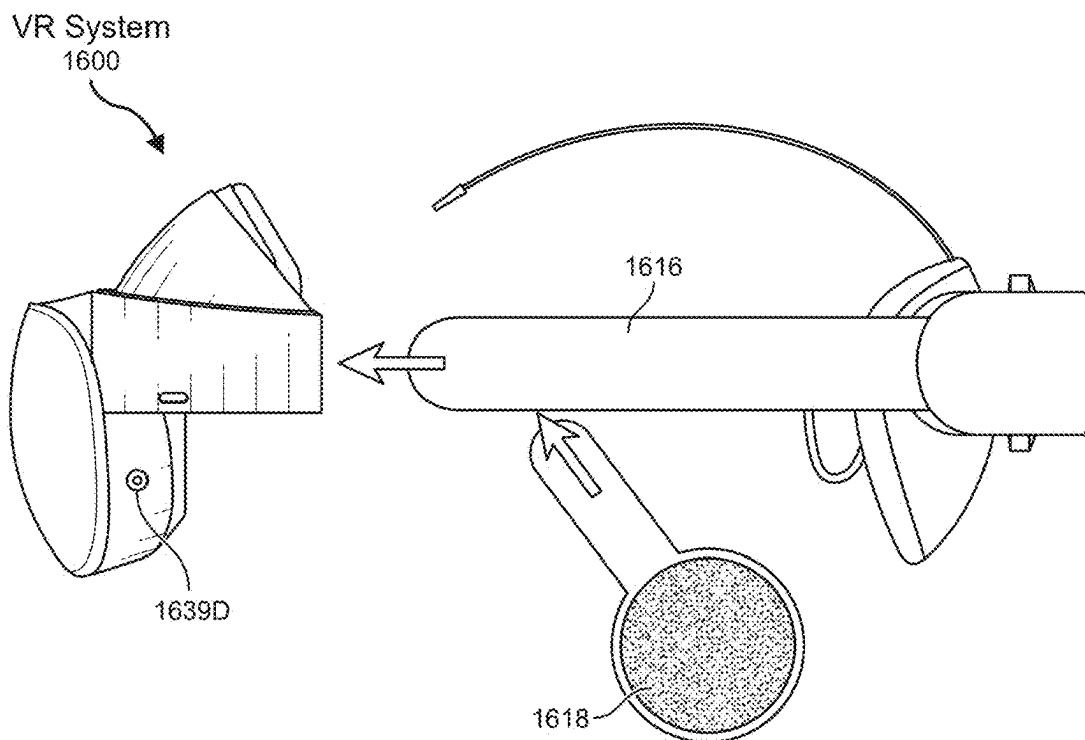


FIG. 16B

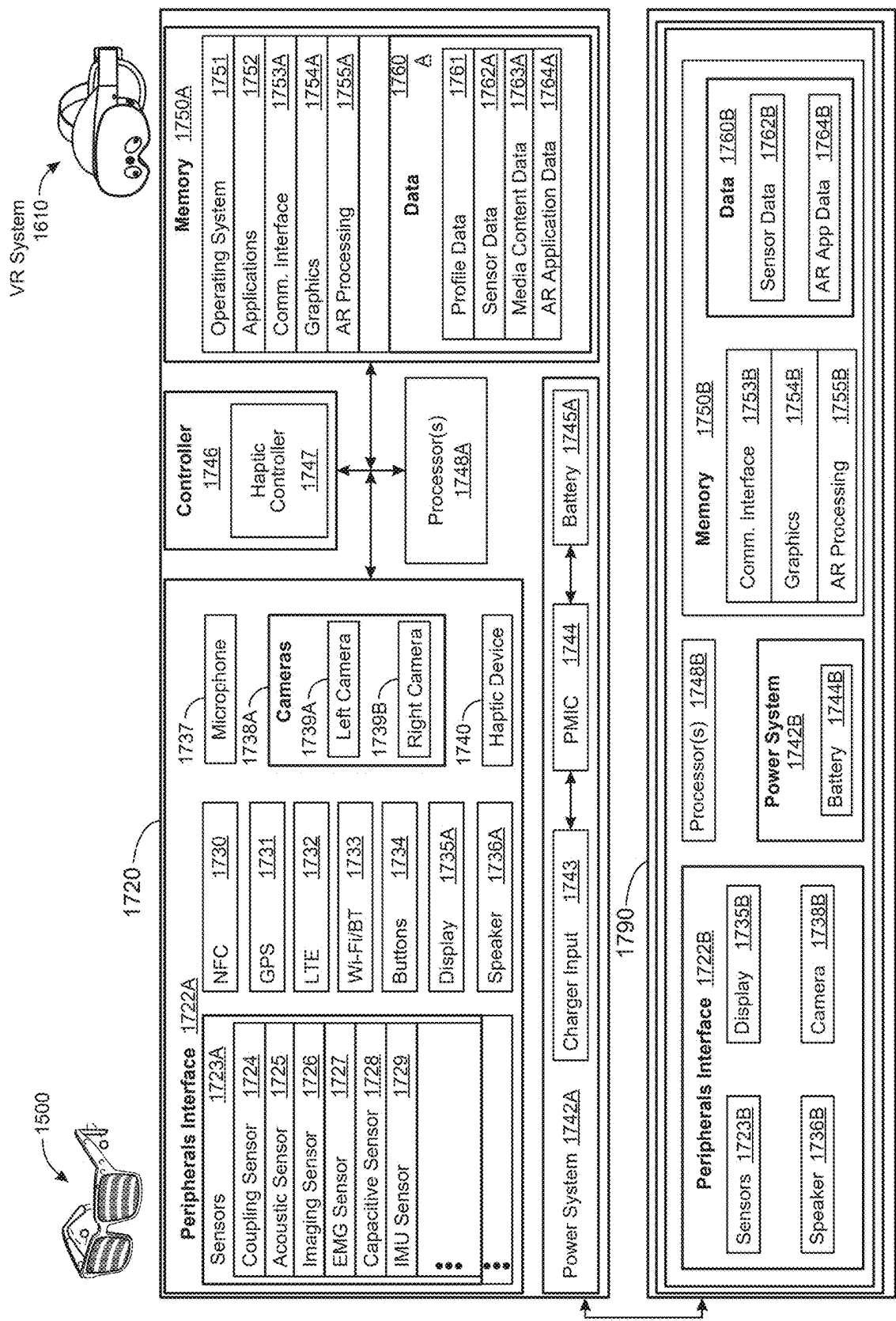


FIG. 17

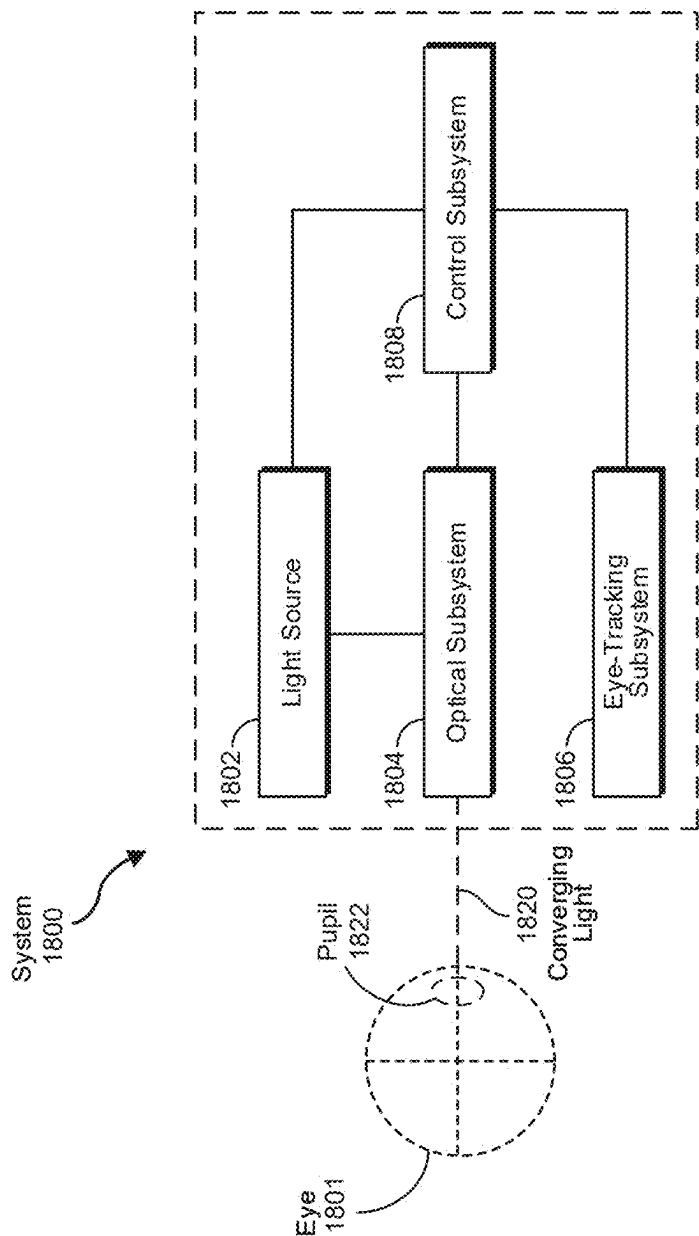


FIG. 18

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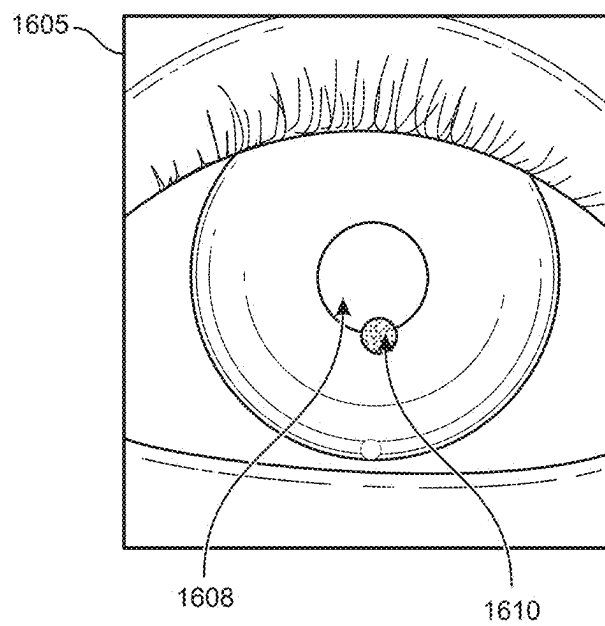
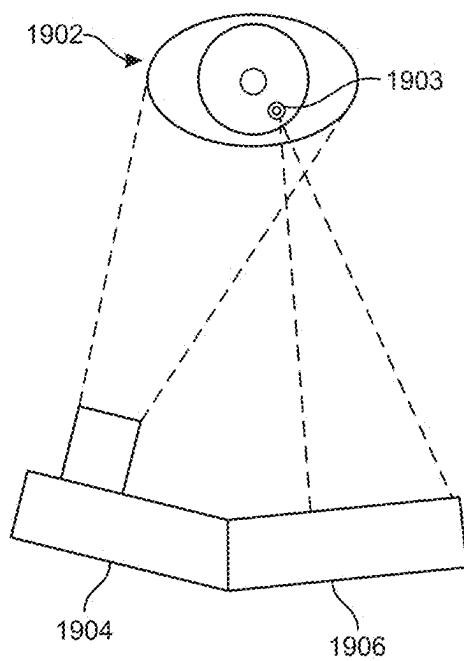



FIG. 19

COMPACT WAVEGUIDE ILLUMINATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application No. 63/560,408, filed Mar. 1, 2024, titled “Compact Waveguide Splitter Circuit,” the disclosure of which is hereby incorporated, in its entirety, by this reference.

BACKGROUND

[0002] Laser-source-based displays often exhibit better color gamut and higher brightness compared to LEDs. Their narrowband nature also enables a wide range of applications with novel optical designs using diffractive/holographic optical elements or metasurfaces.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The accompanying drawings illustrate a number of exemplary embodiments and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.

[0004] FIG. 1 is a diagram of an illumination system according to some embodiments of this disclosure.

[0005] FIG. 2 depicts an illumination system comprising an evanescent waveguide splitter according to some embodiments of this disclosure.

[0006] FIG. 3 depicts a mode combiner waveguide structure according to some embodiments of this disclosure.

[0007] FIG. 4 is a diagram of a dual injection evanescent waveguide splitter according to some embodiments of this disclosure.

[0008] FIG. 5 depicts an illumination system comprising a nested Y-splitter tree according to some embodiments of this disclosure.

[0009] FIG. 6 is a diagram of an illustrative adiabatic directional coupler according to some embodiments of this disclosure.

[0010] FIG. 7 is a diagram of an illustrative waveguide crossing according to some embodiments of this disclosure.

[0011] FIG. 8 depicts an illumination system comprising a star coupler according to some embodiments of this disclosure.

[0012] FIG. 9 is an illustration of an example artificial-reality system according to some embodiments of this disclosure.

[0013] FIG. 10 is an illustration of an example artificial-reality system with a handheld device according to some embodiments of this disclosure.

[0014] FIG. 11A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0015] FIG. 11B is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0016] FIG. 12A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0017] FIG. 12B is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0018] FIG. 13 is an illustration of an example wrist-wearable device of an artificial-reality system according to some embodiments of this disclosure.

[0019] FIG. 14 is an illustration of an example wearable artificial-reality system according to some embodiments of this disclosure.

[0020] FIG. 15 is an illustration of an example augmented-reality system according to some embodiments of this disclosure.

[0021] FIG. 16A is an illustration of an example virtual-reality system according to some embodiments of this disclosure.

[0022] FIG. 16B is an illustration of another perspective of the virtual-reality systems shown in FIG. 16A.

[0023] FIG. 17 is a block diagram showing system components of example artificial- and virtual-reality systems.

[0024] FIG. 18 is an illustration of an example system that incorporates an eye-tracking subsystem capable of tracking a user's eye(s).

[0025] FIG. 19 is a more detailed illustration of various aspects of the eye-tracking subsystem illustrated in FIG. 18.

[0026] Throughout the drawings, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the exemplary embodiments described herein are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. However, the exemplary embodiments described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within the scope of the appended claims.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0027] Laser-source-based displays often exhibit better color gamut and higher brightness compared to LEDs. For example, a photonic integrated circuit (PIC)-based front/backlight illuminator may offer advantages such as low power consumption, structural compactness, and high controllability of the emitted light with respect to numerical aperture (NA), chief ray angle, polarization as well as emission cone angle. However, a large overhead area is often required for the part of optical dispatching circuit compared to the active display area, which may significantly increase the fabrication cost of PIC devices.

[0028] The present application is directed to optical dispatching circuits that may reduce the footprint of an illumination system. In particular, embodiments of the present application provide an illumination system that splits and spreads incoming light sources (e.g., RGB laser light sources) into a plurality of emitters that cover a two-dimensional (2D) area. In some embodiments, the light sources may be split and spread across the emitters with a low insertion loss and/or a high uniformity. The present application describes various implementations of an optical dispatching circuit, which receives light as input and is configured to spread this light across a number of waveguides that each emit light from a plurality of locations. The optical dispatching circuits described herein may be configured to receive light from multiple sources emitting at different wavelengths (such as, but not limited to, red, green and blue light) and effectively deliver the light from the

multiple sources in a substantially uniform manner to a plurality of emitters that cover a 2D area.

[0029] FIG. 1 is a diagram of an illumination system according to some embodiments of this disclosure. In the example of FIG. 1, illumination system 100 comprises a display area 112 for emitting light and an optical dispatching circuit 120, which receives light inputs 110a, 110b and 110c. The illumination system 100 may for instance, be part of a display for a device such as a head-mounted unit, AR/VR or other artificial reality, mixed reality, augmented reality, extended reality, device, mobile device, etc.

[0030] In the example of FIG. 1, the display area 112 comprises a plurality of waveguides 114 arranged on a substrate (e.g., transparent glass or quartz substrate). The waveguides in the example of FIG. 1 are arranged in groups of three waveguides, one for each of the colors of light input as inputs 110a, 110b and 110c. The light inputs 110a, 110b and 110c may, for instance, provide red, green and blue light, respectively, and the optical dispatching circuit is configured to direct light from these inputs to one of the waveguides in each group of three waveguides. As such, in this example for a given group of three waveguides, red light would propagate through one of the waveguides, green light would propagate through another one of the waveguides, and blue light would propagate through the other waveguide. As illustrated in FIG. 1, the waveguides 114 may be arranged in a particular sequence (e.g., RGB) which may repeat such that rows 116 may be established across the waveguide columns 114. Any number of waveguide rows and columns may be provided in the illumination system 100, and any number of emitters may be provided in each waveguide.

[0031] In the example of FIG. 1, each of the waveguides 114 comprises a plurality of emitters as shown in insert 118 which represents the region 117. The emitters in this example are grating emitters, and may be provided along the length of each of the waveguides. As such, the emitters may be arranged over the display area 112 and configured to emit light across this area when light propagates through the respect waveguide. A cross-sectional view through three of the waveguides 114 is shown in inset 119, illustrating that the waveguides are arranged as structures on top of a substrate 121.

[0032] In summary, the light input to the optical dispatching circuit 120 is distributed, by the optical dispatching circuit, to the waveguides 114, with the input light 110a being distributed to a first set of waveguides, the input light 110b being distributed to a second set of waveguides and the input light 110c being distributed to a third set of waveguides of the display area 112. Emitter structures are formed in each of the waveguides 114 that produce light from the waveguides perpendicular to the light propagation direction in the waveguide. In the example of FIG. 1, the light propagates along the y-axis in the waveguides, and is emitted in a direction coming out of the page (z-axis).

[0033] In some embodiments, the light inputs 110a, 110b and 110c each comprise a laser light source. Additionally, or alternatively, the light inputs 110a, 110b and 110c may comprise a respective waveguide that directs light into the optical dispatching circuit 120, and which may receive light from a light source separate from the illumination system 100 (e.g., laser sources are connected to waveguides of the illumination system 100, which provide light inputs to the optical addressing circuit).

[0034] In some embodiments, a width of each of the waveguides 114 (e.g., in the example of FIG. 1 the size of the waveguide in the x-direction) is greater than or equal to 1 μm , 1.5 μm , 2 μm , 2.5 μm or 3 μm . In some embodiments, the width of each of the waveguides 114 is less than or equal to 3.5 μm , 3 μm , 2.5 μm , 2 μm or 1.5 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of each of the waveguides 114 is greater or equal to 1.5 μm and less than or equal to 2.5 μm , etc.).

[0035] To produce a display, the illumination system 100 may be combined with a suitable panel that is configured to allow the light from the illumination system to pass through, or be blocked by, individual pixels. For instance, a display may comprise the illumination system 100 arranged next to a liquid crystal on silicon (LCOS) panel. Light from the emitters of the illumination system 100 may be incident on any number of pixels in a display panel, including a large number of pixels (e.g., hundreds). That is, the emitters of the illumination system 100 may not be pixels, but rather emitters that produce diverging light rays that together cover the surface of a display panel. In such a configuration, the global uniformity of the light produced may be more important than the relative brightness of light produced by the emitters. That is, if some emitters emit more light than other emitters, this may not present an issue so long as the light combined from all the emitters produces a suitably uniform light field.

[0036] Hereinafter, the three light inputs 110a, 110b and 110c to the optical dispatching circuit 120 will be described as being red, green and blue light for purposes of illustration. However, it will be appreciated that these three light inputs may in general be configured to accept light of any three different wavelengths, and the techniques described herein are not necessarily limited to the use of any particular wavelengths or colors of light.

[0037] Described below are various approaches for implementing the optical dispatching circuit 120. Each of these approaches may be utilized in the system of FIG. 1 or in any other suitable system in which light from multiple sources are directed to a plurality of waveguides, with light from a given source being directed to one subset of the waveguides.

[0038] FIG. 2 depicts an illumination system comprising an evanescent waveguide splitter according to some embodiments of this disclosure. In the example of FIG. 2, the optical dispatching circuit 120 comprises bus waveguide 211 and transport slab 212. Light 110, which may include light from light inputs 110a, 110b and 110c combined together, is input to the bus waveguide 211 and propagates through the bus waveguide. According to some embodiments, the bus waveguide may be implemented as a photonic integrated circuit (PIC). The light in the bus waveguide couples across the gap to the transport slab 212, which is arranged adjacent to and apart from the bus waveguide, and as a result the bus waveguide injects the different wavelengths of light together into the transport slab. The light propagates through the transport slab and into a plurality of waveguide receivers that include the labeled waveguide receivers 214a and 214d, which propagate the light into respective coarse-wavelength multiplexers (CWMs) 220. In the implementation of optical dispatching circuit 120 shown in FIG. 2, the CWMs 220 split the light into different frequency components, which propagate through three distinct waveguides coupled to a given CWM. Slab 210 is optional and may not be included in all

implementations. In some examples, to further reduce footprint, CWMs 220 may be a directional coupler or an inverse designed pattern.

[0039] As shown in FIG. 2, the gap between the bus waveguide 211 and the transport slab 212 narrows with distance (along the y-direction) from the input to the optical dispatching circuit 120. For instance, the gap between the bus waveguide 211 and the transport slab 212 proximate to receiver 214a is smaller than the gap between the waveguides proximate to receiver 214d. This configuration may provide for a relative constant amount of light propagating into each of the different waveguides receivers. For instance, the amount of light propagating through the bus waveguide 211 will generally decrease with increasing distance along the bus waveguide in the y-direction. As a result, with a constant gap between the bus waveguide 211 and the transport slab 212, the amount of light that would propagate through receiver 214a would be less than receiver 214d. By reducing the gap between the waveguides, the fraction of the light in the bus waveguide that leaks into the transport slab increases. As such, by tuning the size of the gap along the length of the bus waveguide, a uniform transfer of light into the different receivers may be achieved.

[0040] According to some embodiments, different wavelengths of light may leak from the bus waveguide 211 into the transport slab 212 at different rates. As such, tuning the size of the gap along the bus waveguide may not be sufficient to direct a uniform amount of light into each waveguide, because some waveguides may receive more light at certain wavelengths if those wavelengths more readily leak from the bus waveguide into the transport slab than other wavelengths. For instance, red light may be expected to leak from the bus waveguide into the transport slab at a higher rate than blue light. The inventors have recognized that light 110 may be configured with different frequencies of light provided in different modes, which provide for substantially uniform coupling of the different frequencies from the bus waveguide into the transport slab. For example, light 110 may comprise red light in a first mode (e.g., the zeroth fundamental mode), green light in a second mode (e.g., the first fundamental mode), and blue light in a third mode (e.g., the second fundamental mode), where the first, second and third modes are different modes of light. According to some embodiments, light 110 may comprise red light in a fundamental 0th mode, green light in a 1st higher-order mode, and blue light in a 2nd higher-order mode. The optical dispatching circuit 120 as shown in FIG. 2 may advantageously exhibit significant footprint reduction compared to other optical circuit designs.

[0041] According to some embodiments, optical dispatching circuit 120 shown in FIG. 2 may include an additional CWM coupled to the bus waveguide 211 and configured to input the light 110 into the bus waveguide. The additional CWM (not shown in FIG. 2) may be configured to receive multiple wavelengths of light and to adjust the modes of these wavelengths to produce different frequencies of light provided in different modes, as described above.

[0042] According to some embodiments, an additional CWM, whether included in optical dispatching circuit 120 or provided separately, may create different modes of light that are input into the bus waveguide 211 may comprise an asymmetric directional coupler. An example of such a CWM is shown in FIG. 3.

[0043] In the example of FIG. 3, CWM 300 comprises three waveguides 311, 312 and 313. Multiple lasers may be separately coupled to the waveguides 311, 312 and 313 in the fundamental mode. For instance, the CWM 300 may comprise, or may be coupled to, a red laser, a green laser, and a blue laser which direct red, green and blue light, respectively, in the fundamental mode, into the waveguides 311, 312 and 313, respectively.

[0044] The illustrative CWM 300 is configured to combine light from three light sources of different wavelengths, propagating in separate waveguides, into a single waveguide where the light exhibits three different modes. In the example of FIG. 3, green light is propagating in the waveguide 311, red light in the waveguide 312, and blue light in the waveguide 313. In the section 301a, shown in more detail as inset 301b, the green light in waveguide 311 is combined with the red light in the waveguide 312 through a directional coupler. The directional coupler may be designed with two close waveguides of different widths, and the effective indices of the waveguides 311 and 312 may be carefully tuned, such that the green light is transferred from the fundamental mode of waveguide 311 to the 1st higher order mode in waveguide 312 (e.g., with near-unity conversion efficiency). The Red light may remain the fundamental mode in waveguide 312. Similarly, as in the section 302a, shown in more detail as inset 302b, blue light in waveguide 313 may be transferred from the fundamental mode of waveguide 313 to the 2nd higher-order mode in waveguide 312, while the red and green light in the waveguide 312 remain in the fundamental and 1st higher order modes, respectively. As a result, the CWM may produce red, green and blue light together, in different modes. This light may be provided as input light 110 to the bus waveguide 211 as shown in FIG. 2 and described above.

[0045] According to some embodiments, any one or more of the waveguides 311, 312 and 313 may be tapered in width in one or both of the sections 301a and 302a to produce a desired transfer of the light from one of waveguides 311 or 313 into the waveguide 312. For example, in section 301a, the waveguide 311 may be tapered to be decreasing in width from left to right as shown in FIG. 3, while the waveguide 312 may optionally be tapered to be increasing in width from left to right. By tuning the change in width and size of the gap between the waveguides 311 and 312, the fundamental mode of light in the waveguide 311 may couple to the 1st higher order mode of light in the waveguide 312 with a very high (e.g., near unity) transition efficiency. Similarly, in section 302a, the waveguide 313 may be tapered to be decreasing in width from left to right as shown in FIG. 3, while the waveguide 312 may optionally be tapered to be increasing in width from left to right. By tuning the change in width and size of the gap between the waveguides 313 and 312, the fundamental mode of light in the waveguide 313 may couple to the 2nd higher order mode of light in the waveguide 312. It may be noted in the example of FIG. 3 that the waveguides 311, 312 and 313 do not touch, but rather come close to each other and taper narrower in the sections 301a and 302a.

[0046] In some embodiments, a width of waveguide 311 is greater than or equal to 350 nm, 400 nm, 450 nm, 500 nm, 550 nm or 600 nm. In some embodiments, the width of waveguide 311 is less than or equal to 650 nm, 600 nm, 550 nm, 500 nm, 450 nm, or 400 nm. Any suitable combinations of the above-referenced ranges are also possible (e.g., the

width of waveguide **311** is greater or equal to 400 nm and less than or equal to 600 nm, etc.). The width of the waveguide **311** may, in some embodiments, narrow along the section **301a** as described above. For instance, the width may narrow from 550 nm to 450 nm.

[0047] In some embodiments, a width of waveguide **313** is greater than or equal to 200 nm, 250 nm, 300 nm, 350 nm, or 400 nm. In some embodiments, the width of waveguide **313** is less than or equal to 450 nm, 400 nm, 350 nm, 300 nm, or 250 nm. Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of waveguide **313** is greater or equal to 250 nm and less than or equal to 400 nm, etc.). The width of the waveguide **313** may, in some embodiments, narrow along the section **301b** as described above. For instance, the width may narrow from 350 nm to 270 nm.

[0048] In some embodiments, a width of waveguide **312** is greater than or equal to 900 nm, 1 μm , 1.1 μm , 1.2 μm , 1.3 μm or 1.4 μm . In some embodiments, the width of waveguide **312** is less than or equal to 1.5 μm , 1.4 μm , 1.3 μm , 1.2 μm , 1.1 μm or 1 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of waveguide **312** is greater or equal to 1.1 μm and less than or equal to 1.3 μm , etc.). The width of the waveguide **312** may, in some embodiments, widen along the section **301a** or section **301b** as described above. For instance, the width may widen from 1.15 μm to 1.25 μm .

[0049] In the example of FIG. 2, the amount of light emitted from the emitters of the waveguides **114** in display area **112** may decrease along the x-direction due to the amount of light in the waveguides gradually decreasing as more and more is emitted from the waveguide through the emitters. In some embodiments, the optical dispatching circuit **120** may be configured with multiple instances of the waveguides shown in FIG. 2, where the receivers of the optical dispatching circuit **120** are arranged at opposing ends of the waveguides **114** in the display area **112**.

[0050] FIG. 4 is a diagram of a dual injection evanescent waveguide splitter that utilizes this approach, according to some embodiments of this disclosure. In the example of FIG. 4, the optical dispatching circuit **120** comprises two instances of the bus waveguide **211** and transport slab **212**, in addition to two corresponding sets of receivers and CWMs **220**. By injecting light into the waveguides **114** in display area **112** at both ends, the emitters in the display area **112** may more evenly produce light.

[0051] FIG. 5 depicts an illumination system comprising a nested Y-splitter tree according to some embodiments of this disclosure. FIG. 5 depicts another illustrative design for optical dispatching circuit **120**. In the example of FIG. 5, the optical dispatching circuit **120** comprises a plurality of waveguides arranged in a tree structure. As shown, the tree **501** receives light **110a**, **110b** and **110c** into three separate waveguides, which each split into two branches at a Y-split structure. The tree **501** also includes a plurality of waveguide crossers **511**, shown as rectangles in the tree (only one is labeled in FIG. 5).

[0052] In the example of FIG. 5, each waveguide crosser **511** couples to two waveguides as inputs and two waveguides as outputs. In operation, the waveguide crosser receives light of two different wavelengths along the two input waveguides, and outputs the received light from the two output waveguides, with the light inputs spatially swapped. That is, the light provided to the left side input of

a waveguide crosser may be output from the right side output of the waveguide crosser, and vice versa. The waveguide crossers allow for the light inputs **110a**, **110b** and **110c** at each wavelength to be output from a plurality of waveguides at the other end of the tree **501** by allowing the paths of a given wavelength of light to effectively pass through or over the paths of a different wavelength of light. For example, the labeled waveguide crosser **511** has light **110b** (e.g., green) and light **110a** (e.g., red) as inputs on the left and the right, respectively. This waveguide crosser **511** then outputs red light from the left output and green light from the right output, effectively crossing the red and green light paths.

[0053] As shown in the example of FIG. 5, the tree **501** outputs are a plurality of waveguides arranged in a series with a set of each input color being arranged as a group next to each other. For example, as shown in FIG. 5, the output waveguides propagate light in the following order from left to right: blue, green, red, blue, green, red, blue, green, red, blue, green and red. The tree **501** may be arranged with any number of output waveguides by selecting an appropriate number of waveguide Y-split structures and waveguide crossers.

[0054] In the example of FIG. 5, irrespective of the number of waveguides arranged at the output of tree **501**, the light from these waveguides is directed through an array of waveguides **502** into the display area **112**, which produces light from the emitters along the waveguides as shown and as described above. The optical dispatching circuit **120** shown in FIG. 5 may provide for highly uniform light across the rows of waveguides **114** in the display area **112** since the input light may be evenly split at the Y-split structures and crossed without significant loss in the waveguide crossers **511**.

[0055] FIG. 6 is a diagram of an illustrative directional coupler that may be suitable for use as a waveguide crosser **511**, according to some embodiments of this disclosure. In the example of FIG. 6, the directional coupler **600** is configured to swap the spatial modes of light provided at the two inputs to the directional coupler. As shown, directional coupler **600** is left-right symmetrical and includes two waveguides **601** and **602** that taper in width from one side to the other. In particular, waveguide **601** widens from the input to the output side, and waveguide **602** narrows from the input to the output side. The widths of the waveguides at the input side may be different from one another, and the widths of the waveguides at the output side may be different from one another.

[0056] According to some embodiments, the dimensions of the waveguides **601** and **602** may be tuned so that the light input to waveguide **601** (**602**) will be transmitted to waveguide **602** (**601**). While different wavelengths of light may be transmitted in a different manner, the inventors have recognized that with a sufficiently long coupler and thus a sufficiently slow tapered waveguide guide, the directional coupler will produce high (e.g., close to 100%) power transfer of light that is independent of light frequency. In some embodiments, the length of the tapered section of the waveguides **608** is between 500 μm and 750 μm . The gap between the waveguides may be between 400 nm and 600 nm.

[0057] FIG. 7 is a diagram of an illustrative waveguide crossing that may be suitable for use as waveguide crosser **511**, according to some embodiments of this disclosure. A

more compact waveguide crossing may be realized through an inverse design with the technique of shape optimization of the waveguide. In the example of FIG. 7, inverse-designed waveguide crossing **700** is a waveguide shaped as a crossing between two inputs and two outputs. For example, light provided to input **701** through the left may be routed to the output **703**, whereas light provided to input **702** may be routed to the output **704**.

[0058] FIG. 8 depicts an illumination system comprising a star coupler according to some embodiments of this disclosure. FIG. 8 depicts another illustrative design for optical dispatching circuit **120**. In the example of FIG. 8, light **110a**, **110b** and **110c** is input to three separate waveguides **801**, **802** and **803**, respectively, which terminate with tips **804** that are arranged in close proximity to one another. Light from the waveguides **801**, **802** and **803** propagates into the slab waveguide **805** and spreads out radially in the slab waveguide. The light from the slab waveguide is then collected by a large array of receiver waveguides arranged at the opposing side of the tips **804**. The receiver waveguides include illustrative waveguides **806** and **807**, and the receiver waveguides collect light of each wavelength represented by light **110a**, **110b** and **110c**. The receiver waveguides are coupled via an array of waveguides **810** to waveguides **811**, which are coupled to CWMs **820**. The CWMs **820** split the incoming light into the wavelengths of the input light and direct this light into the waveguides of the display area **112**. The number of receiver waveguides is the same as the number of waveguides **811** and CWMs **820**.

[0059] As shown in FIG. 8, the receiver waveguides are arranged to be wider the further from the central axis of the slab waveguide **805** they are located. Since the light propagates from the tips **804** into the slab waveguide **805**, more light will propagate in the center of the slab waveguide, and less light will propagate the further outward from the central axis the receiver waveguide is placed (e.g., the received light power may be Gaussian with angle of propagation from the tip to the receiver waveguides). As such, the receiver waveguides may be gradually larger so that each receiver waveguide collects substantially the same amount of light, and consequently the waveguides **114** of display area **112** receive substantially the same amount of light as one another.

[0060] In some embodiments, a width of the waveguides **801**, **802** and **803** at the tip ends **804** is greater than or equal to 0.8 μm , 0.9 μm , 1 μm , 1.1 μm , or 1.2 μm . In some embodiments, the width of the waveguides **801**, **802** and **803** at the tip ends **804** is less than or equal to 1.3 μm , 1.2 μm , 1.1 μm , 1 μm , or 0.9 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of the waveguides **801**, **802** and **803** at the tip ends **804** is greater or equal to 0.9 μm and less than or equal to 1.1 μm , etc.). In some embodiments, the waveguides **801**, **802** and **803** widen at the tip ends **804** compared with the width of the waveguides leading up to the tip ends **804** (e.g., the waveguides are 0.7 μm wide and widen to 1 μm wide at the tips).

[0061] In some embodiments, the slab waveguide **805** is a 0.5 mm to 2 mm long waveguide shaped as an arc or annular arc (e.g., a half-circle or half-circle annulus).

[0062] Features from any of the embodiments described herein may be used in combination with one another in accordance with the general principles described herein. These and other embodiments, features, and advantages will

be more fully understood upon reading the following detailed description in conjunction with the accompanying drawings and claims.

EXAMPLE EMBODIMENTS

[0063] Example 1. A device comprising: a plurality of waveguides consisting of a first plurality of waveguides, a second plurality of waveguides and a third plurality of waveguides; and an optical dispatching circuit comprising a first input, a second input and a third input, the optical dispatching circuit configured to direct light from the first input through each of the first plurality of waveguides, to direct light from the second input through each of the second plurality of waveguides, and to direct light from the third input through each of the third plurality of waveguides.

[0064] Example 2. The device of example 1, wherein the optical dispatching circuit comprises: an input multiplexer coupled to the first input, the second input and the third input and comprising an output; a bus waveguide coupled to the output of the input multiplexer; and a transport slab arranged adjacent to and apart from the bus waveguide, the transport slab coupled to each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

[0065] Example 3. The device of example 2, wherein a distance between the bus waveguide and the transport slab narrows from a first end of the bus waveguide proximate to the input multiplexer to an opposing end of the bus waveguide distal from the input multiplexer.

[0066] Example 4. The device of example 2, further comprising a plurality of output multiplexers that couple the transport slab to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

[0067] Example 5. The device of example 2, wherein the input multiplexer comprises: a first multiplexer waveguide coupled to the first input; a second multiplexer waveguide coupled to the second input; and a third multiplexer waveguide coupled to the third input and coupled to the output of the input multiplexer, wherein the first multiplexer waveguide widens from the first input to the output of the input multiplexer, wherein the second multiplexer waveguide narrows from the second input to the output of the input multiplexer, and wherein the third multiplexer waveguide narrows from the third input to the output of the input multiplexer.

[0068] Example 6. The device of example 1, wherein the optical dispatching circuit comprises a waveguide tree coupled to the first input, the second input and the third input, and comprising a plurality of outputs that includes an output for each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

[0069] Example 7. The device of example 6, wherein the waveguide tree comprises a plurality of Y-shaped waveguide sections.

[0070] Example 8. The device of example 6, wherein the waveguide tree comprises a plurality of directional couplers, each directional coupler configured to swap spatial modes of light across a pair of waveguides therein.

[0071] Example 9. The device of example 8, wherein, in each directional coupler of the plurality of directional couplers, one of the pair of waveguides tapers narrower across the directional coupler from an input side to an output side while another of the pair of waveguides tapers wider from the input side to the output side of the directional coupler.

[0072] Example 10. The device of example 1, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

[0073] Example 11. The device of example 10, further comprising a plurality of output multiplexers that couple each of the plurality of receiving waveguides to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each receiving waveguide of the plurality of receiving waveguides is coupled to a respective output multiplexer of the plurality of output multiplexers, and wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

[0074] Example 12. The device of example 10, wherein a width of each receiving waveguide of the plurality of receiving waveguides increases with distance from a central axis of the coupler waveguide.

[0075] Example 13. The device of example 10, further comprising: a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide, wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

[0076] Example 14. The device of example 1, further comprising: a first light source arranged to direct light of a first wavelength through the first input of the optical dispatching circuit; a second light source arranged to direct light of a second wavelength, different to the first wavelength, through the second input of the optical dispatching circuit; and a third light source arranged to direct light of a third wavelength, different to the first wavelength and the second wavelength, through the third input of the optical dispatching circuit.

[0077] Example 15. The device of example 14, wherein the first wavelength is between 600 nm and 700 nm, the second wavelength is between 500 nm and 600 nm, and the third wavelength is between 400 nm and 500 nm.

[0078] Example 16. The device of example 1, wherein each of the plurality of waveguides comprises a plurality of emitter structures configured to output light.

[0079] Example 17. A display comprising the device of example 16 and a liquid crystal on silicon (LCoS) panel arranged to receive light output from the plurality of emitter structures.

[0080] Example 18. The display of example 17 wherein the LCoS panel comprises a plurality of pixels each arranged to receive light from a plurality of the emitter structures.

[0081] Example 19. A method comprising: directing light from a first light source into a first input of an optical

dispatching circuit, through the optical dispatching circuit, into a first plurality of waveguides and into a first plurality of emitter structures arranged within the first plurality of waveguides; directing light from a second light source into a second input of the optical dispatching circuit, through the optical dispatching circuit, into a second plurality of waveguides and into a second plurality of emitter structures arranged within the second plurality of waveguides; and directing light from a third light source into a third input of the optical dispatching circuit, through the optical dispatching circuit, into a third plurality of waveguides and into a third plurality of emitter structures arranged within the third plurality of waveguides.

[0082] Example 20. The method of example 19, wherein the first light source produces light having a wavelength between 600 nm and 700 nm, wherein the second light source produces light having a wavelength between 500 nm and 600 nm, and wherein the third light source produces light having a wavelength between 400 nm and 500 nm.

[0083] Example 21. The method of example 19, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

[0084] Example 22. The method of example 21, wherein: directing light from the first light source into the first input of the optical dispatching circuit comprises directing the light from the first light source through a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; directing light from the second light source into the second input of the optical dispatching circuit comprises directing the light from the second light source through a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; and directing light from the third light source into the third input of the optical dispatching circuit comprises directing the light from the third light source through a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide.

[0085] Example 23. The method of example 22, wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

[0086] Embodiments of the present disclosure may include or be implemented in conjunction with various types of Artificial-Reality (AR) systems. AR may be any super-imposed functionality and/or sensory-detectable content presented by an artificial-reality system within a user's physical surroundings. In other words, AR is a form of reality that has been adjusted in some manner before presentation to a user. AR can include and/or represent virtual reality (VR), augmented reality, mixed AR (MAR), or some combination and/or variation of these types of realities. Similarly, AR environments may include VR environments (including non-immersive, semi-immersive, and fully immersive VR environments), augmented-reality environments (including marker-based augmented-reality environments, markerless augmented-reality environments, location-based augmented-reality environments, and projection-based augmented-reality environments), hybrid-reality environments, and/or any other type or form of mixed- or alternative-reality environments.

[0087] AR content may include completely computer-generated content or computer-generated content combined with captured (e.g., real-world) content. Such AR content may include video, audio, haptic feedback, or some combination thereof, any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional (3D) effect to the viewer). Additionally, in some embodiments, AR may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, for example, create content in an artificial reality and/or are otherwise used in (e.g., to perform activities in) an artificial reality.

[0088] AR systems may be implemented in a variety of different form factors and configurations. Some AR systems may be designed to work without near-eye displays (NEDs). Other AR systems may include a NED that also provides visibility into the real world (such as, e.g., augmented-reality system 1500 in FIG. 15) or that visually immerses a user in an artificial reality (such as, e.g., virtual-reality system 1600 in FIGS. 16A and 16B). While some AR devices may be self-contained systems, other AR devices may communicate and/or coordinate with external devices to provide an AR experience to a user. Examples of such external devices include handheld controllers, mobile devices, desktop computers, devices worn by a user, devices worn by one or more other users, and/or any other suitable external system.

[0089] FIGS. 9-12B illustrate example artificial-reality (AR) systems in accordance with some embodiments. FIG. 9 shows a first AR system 900 and first example user interactions using a wrist-wearable device 902, a head-wearable device (e.g., AR glasses 1500), and/or a handheld intermediary processing device (HIPD) 906. FIG. 10 shows a second AR system 1000 and second example user interactions using a wrist-wearable device 1002, AR glasses 1004, and/or an HIPD 1006. FIGS. 11A and 11B show a third AR system 1100 and third example user interactions using a wrist-wearable device 1102, a head-wearable device (e.g., VR headset 1150), and/or an HIPD 1106. FIGS. 12A and 12B show a fourth AR system 1200 and fourth example user interactions using a wrist-wearable device 1230, VR headset 1220, and/or a haptic device 1260 (e.g., wearable gloves).

[0090] A wrist-wearable device 1300, which can be used for wrist-wearable device 902, 1002, 1102, 1230, and one or more of its components, are described below in reference to FIGS. 13 and 14; head-wearable devices 1500 and 1600, which can respectively be used for AR glasses 904, 1004 or VR headset 1150, 1220, and their one or more components are described below in reference to FIGS. 15-17.

[0091] Referring to FIG. 9, wrist-wearable device 902, AR glasses 904, and/or HIPD 906 can communicatively couple via a network 925 (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.). Additionally, wrist-wearable device 902, AR glasses 904, and/or HIPD 906 can also communicatively couple with one or more servers 930, computers 940 (e.g., laptops, computers, etc.), mobile devices 950 (e.g., smartphones, tablets, etc.), and/or other electronic devices via network 925 (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.).

[0092] In FIG. 9, a user 908 is shown wearing wrist-wearable device 902 and AR glasses 904 and having HIPD 906 on their desk. The wrist-wearable device 902, AR glasses 904, and HIPD 906 facilitate user interaction with an AR environment. In particular, as shown by first AR system

900, wrist-wearable device 902, AR glasses 904, and/or HIPD 906 cause presentation of one or more avatars 910, digital representations of contacts 912, and virtual objects 914. As discussed below, user 908 can interact with one or more avatars 910, digital representations of contacts 912, and virtual objects 914 via wrist-wearable device 902, AR glasses 904, and/or HIPD 906.

[0093] User 908 can use any of wrist-wearable device 902, AR glasses 904, and/or HIPD 906 to provide user inputs. For example, user 908 can perform one or more hand gestures that are detected by wrist-wearable device 902 (e.g., using one or more EMG sensors and/or IMUs, described below in reference to FIGS. 13 and 14) and/or AR glasses 904 (e.g., using one or more image sensor or camera, described below in reference to FIGS. 15-10) to provide a user input. Alternatively, or additionally, user 908 can provide a user input via one or more touch surfaces of wrist-wearable device 902, AR glasses 904, HIPD 906, and/or voice commands captured by a microphone of wrist-wearable device 902, AR glasses 904, and/or HIPD 906. In some embodiments, wrist-wearable device 902, AR glasses 904, and/or HIPD 906 include a digital assistant to help user 908 in providing a user input (e.g., completing a sequence of operations, suggesting different operations or commands, providing reminders, confirming a command, etc.). In some embodiments, user 908 can provide a user input via one or more facial gestures and/or facial expressions. For example, cameras of wrist-wearable device 902, AR glasses 904, and/or HIPD 906 can track eyes of user 908 for navigating a user interface.

[0094] Wrist-wearable device 902, AR glasses 904, and/or HIPD 906 can operate alone or in conjunction to allow user 908 to interact with the AR environment. In some embodiments, HIPD 906 is configured to operate as a central hub or control center for the wrist-wearable device 902, AR glasses 904, and/or another communicatively coupled device. For example, user 908 can provide an input to interact with the AR environment at any of wrist-wearable device 902, AR glasses 904, and/or HIPD 906, and HIPD 906 can identify one or more back-end and front-end tasks to cause the performance of the requested interaction and distribute instructions to cause the performance of the one or more back-end and front-end tasks at wrist-wearable device 902, AR glasses 904, and/or HIPD 906. In some embodiments, a back-end task is a background processing task that is not perceptible by the user (e.g., rendering content, decompression, compression, etc.), and a front-end task is a user-facing task that is perceptible to the user (e.g., presenting information to the user, providing feedback to the user, etc.). HIPD 906 can perform the back-end tasks and provide wrist-wearable device 902 and/or AR glasses 904 operational data corresponding to the performed back-end tasks such that wrist-wearable device 902 and/or AR glasses 904 can perform the front-end tasks. In this way, HIPD 906, which has more computational resources and greater thermal headroom than wrist-wearable device 902 and/or AR glasses 904, performs computationally intensive tasks and reduces the computer resource utilization and/or power usage of wrist-wearable device 902 and/or AR glasses 904.

[0095] In the example shown by first AR system 900, HIPD 906 identifies one or more back-end tasks and front-end tasks associated with a user request to initiate an AR video call with one or more other users (represented by avatar 910 and the digital representation of contact 912) and distributes instructions to cause the performance of the one

or more back-end tasks and front-end tasks. In particular, HIPD 906 performs back-end tasks for processing and/or rendering image data (and other data) associated with the AR video call and provides operational data associated with the performed back-end tasks to AR glasses 904 such that the AR glasses 904 perform front-end tasks for presenting the AR video call (e.g., presenting avatar 910 and digital representation of contact 912).

[0096] In some embodiments, HIPD 906 can operate as a focal or anchor point for causing the presentation of information. This allows user 908 to be generally aware of where information is presented. For example, as shown in first AR system 900, avatar 910 and the digital representation of contact 912 are presented above HIPD 906. In particular, HIPD 906 and AR glasses 904 operate in conjunction to determine a location for presenting avatar 910 and the digital representation of contact 912. In some embodiments, information can be presented a predetermined distance from HIPD 906 (e.g., within 5 meters). For example, as shown in first AR system 900, virtual object 914 is presented on the desk some distance from HIPD 906. Similar to the above example, HIPD 906 and AR glasses 904 can operate in conjunction to determine a location for presenting virtual object 914. Alternatively, in some embodiments, presentation of information is not bound by HIPD 906. More specifically, avatar 910, digital representation of contact 912, and virtual object 914 do not have to be presented within a predetermined distance of HIPD 906.

[0097] User inputs provided at wrist-wearable device 902, AR glasses 904, and/or HIPD 906 are coordinated such that the user can use any device to initiate, continue, and/or complete an operation. For example, user 908 can provide a user input to AR glasses 904 to cause AR glasses 904 to present virtual object 914 and, while virtual object 914 is presented by AR glasses 904, user 908 can provide one or more hand gestures via wrist-wearable device 902 to interact and/or manipulate virtual object 914.

[0098] FIG. 10 shows a user 1008 wearing a wrist-wearable device 1002 and AR glasses 1004, and holding an HIPD 1006. In second AR system 1000, the wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 are used to receive and/or provide one or more messages to a contact of user 1008. In particular, wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 detect and coordinate one or more user inputs to initiate a messaging application and prepare a response to a received message via the messaging application.

[0099] In some embodiments, user 1008 initiates, via a user input, an application on wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 that causes the application to initiate on at least one device. For example, in second AR system 1000, user 1008 performs a hand gesture associated with a command for initiating a messaging application (represented by messaging user interface 1016), wrist-wearable device 1002 detects the hand gesture and, based on a determination that user 1008 is wearing AR glasses 1004, causes AR glasses 1004 to present a messaging user interface 1016 of the messaging application. AR glasses 1004 can present messaging user interface 1016 to user 1008 via its display (e.g., as shown by a field of view 1018 of user 1008). In some embodiments, the application is initiated and executed on the device (e.g., wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006) that detects the user input to initiate the application, and the device provides

another device operational data to cause the presentation of the messaging application. For example, wrist-wearable device 1002 can detect the user input to initiate a messaging application, initiate and run the messaging application, and provide operational data to AR glasses 1004 and/or HIPD 1006 to cause presentation of the messaging application. Alternatively, the application can be initiated and executed at a device other than the device that detected the user input. For example, wrist-wearable device 1002 can detect the hand gesture associated with initiating the messaging application and cause HIPD 1006 to run the messaging application and coordinate the presentation of the messaging application.

[0100] Further, user 1008 can provide a user input provided at wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 to continue and/or complete an operation initiated at another device. For example, after initiating the messaging application via wrist-wearable device 1002 and while AR glasses 1004 present messaging user interface 1016, user 1008 can provide an input at HIPD 1006 to prepare a response (e.g., shown by the swipe gesture performed on HIPD 1006). Gestures performed by user 1008 on HIPD 1006 can be provided and/or displayed on another device. For example, a swipe gesture performed on HIPD 1006 is displayed on a virtual keyboard of messaging user interface 1016 displayed by AR glasses 1004.

[0101] In some embodiments, wrist-wearable device 1002, AR glasses 1004, HIPD 1006, and/or any other communicatively coupled device can present one or more notifications to user 1008. The notification can be an indication of a new message, an incoming call, an application update, a status update, etc. User 1008 can select the notification via wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 and can cause presentation of an application or operation associated with the notification on at least one device. For example, user 1008 can receive a notification that a message was received at wrist-wearable device 1002, AR glasses 1004, HIPD 1006, and/or any other communicatively coupled device and can then provide a user input at wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 to review the notification, and the device detecting the user input can cause an application associated with the notification to be initiated and/or presented at wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006.

[0102] While the above example describes coordinated inputs used to interact with a messaging application, user inputs can be coordinated to interact with any number of applications including, but not limited to, gaming applications, social media applications, camera applications, web-based applications, financial applications, etc. For example, AR glasses 1004 can present to user 1008 game application data, and HIPD 1006 can be used as a controller to provide inputs to the game. Similarly, user 1008 can use wrist-wearable device 1002 to initiate a camera of AR glasses 1004, and user 1008 can use wrist-wearable device 1002, AR glasses 1004, and/or HIPD 1006 to manipulate the image capture (e.g., zoom in or out, apply filters, etc.) and capture image data.

[0103] Users may interact with the devices disclosed herein in a variety of ways. For example, as shown in FIGS. 11A and 11B, a user 1108 may interact with an AR system 1100 by donning a VR headset 1150 while holding HIPD 1106 and wearing wrist-wearable device 1102. In this

example, AR system 1100 may enable a user to interact with a game 1110 by swiping their arm. One or more of VR headset 1150, HIPD 1106, and wrist-wearable device 1102 may detect this gesture and, in response, may display a sword strike in game 1110. Similarly, in FIGS. 12A and 12B, a user 1208 may interact with an AR system 1200 by donning a VR headset 1220 while wearing haptic device 1260 and wrist-wearable device 1230. In this example, AR system 1200 may enable a user to interact with a game 1210 by swiping their arm. One or more of VR headset 1220, haptic device 1260, and wrist-wearable device 1230 may detect this gesture and, in response, may display a spell being cast in game 1110.

[0104] Having discussed example AR systems, devices for interacting with such AR systems and other computing systems more generally will now be discussed in greater detail. Some explanations of devices and components that can be included in some or all of the example devices discussed below are explained herein for ease of reference. Certain types of the components described below may be more suitable for a particular set of devices, and less suitable for a different set of devices. But subsequent reference to the components explained here should be considered to be encompassed by the descriptions provided.

[0105] In some embodiments discussed below, example devices and systems, including electronic devices and systems, will be addressed. Such example devices and systems are not intended to be limiting, and one of skill in the art will understand that alternative devices and systems to the example devices and systems described herein may be used to perform the operations and construct the systems and devices that are described herein.

[0106] An electronic device may be a device that uses electrical energy to perform a specific function. An electronic device can be any physical object that contains electronic components such as transistors, resistors, capacitors, diodes, and integrated circuits. Examples of electronic devices include smartphones, laptops, digital cameras, televisions, gaming consoles, and music players, as well as the example electronic devices discussed herein. As described herein, an intermediary electronic device may be a device that sits between two other electronic devices and/or a subset of components of one or more electronic devices and facilitates communication, data processing, and/or data transfer between the respective electronic devices and/or electronic components.

[0107] An integrated circuit may be an electronic device made up of multiple interconnected electronic components such as transistors, resistors, and capacitors. These components may be etched onto a small piece of semiconductor material, such as silicon. Integrated circuits may include analog integrated circuits, digital integrated circuits, mixed signal integrated circuits, and/or any other suitable type or form of integrated circuit. Examples of integrated circuits include application-specific integrated circuits (ASICs), processing units, central processing units (CPUs), co-processors, and accelerators.

[0108] Analog integrated circuits, such as sensors, power management circuits, and operational amplifiers, may process continuous signals and perform analog functions such as amplification, active filtering, demodulation, and mixing. Examples of analog integrated circuits include linear integrated circuits and radio frequency circuits.

[0109] Digital integrated circuits, which may be referred to as logic integrated circuits, may include microprocessors, microcontrollers, memory chips, interfaces, power management circuits, programmable devices, and/or any other suitable type or form of integrated circuit. In some embodiments, examples of integrated circuits include central processing units (CPUs),

[0110] Processing units, such as CPUs, may be electronic components that are responsible for executing instructions and controlling the operation of an electronic device (e.g., a computer). There are various types of processors that may be used interchangeably, or may be specifically required, by embodiments described herein. For example, a processor may be: (i) a general processor designed to perform a wide range of tasks, such as running software applications, managing operating systems, and performing arithmetic and logical operations; (ii) a microcontroller designed for specific tasks such as controlling electronic devices, sensors, and motors; (iii) an accelerator, such as a graphics processing unit (GPU), designed to accelerate the creation and rendering of images, videos, and animations (e.g., virtual-reality animations, such as three-dimensional modeling); (iv) a field-programmable gate array (FPGA) that can be programmed and reconfigured after manufacturing and/or can be customized to perform specific tasks, such as signal processing, cryptography, and machine learning; and/or (v) a digital signal processor (DSP) designed to perform mathematical operations on signals such as audio, video, and radio waves. One or more processors of one or more electronic devices may be used in various embodiments described herein.

[0111] Memory generally refers to electronic components in a computer or electronic device that store data and instructions for the processor to access and manipulate. Examples of memory can include: (i) random access memory (RAM) configured to store data and instructions temporarily; (ii) read-only memory (ROM) configured to store data and instructions permanently (e.g., one or more portions of system firmware, and/or boot loaders) and/or semi-permanently; (iii) flash memory, which can be configured to store data in electronic devices (e.g., USB drives, memory cards, and/or solid-state drives (SSDs)); and/or (iv) cache memory configured to temporarily store frequently accessed data and instructions. Memory, as described herein, can store structured data (e.g., SQL databases, MongoDB databases, GraphQL data, JSON data, etc.). Other examples of data stored in memory can include (i) profile data, including user account data, user settings, and/or other user data stored by the user, (ii) sensor data detected and/or otherwise obtained by one or more sensors, (iii) media content data including stored image data, audio data, documents, and the like, (iv) application data, which can include data collected and/or otherwise obtained and stored during use of an application, and/or any other types of data described herein.

[0112] Controllers may be electronic components that manage and coordinate the operation of other components within an electronic device (e.g., controlling inputs, processing data, and/or generating outputs). Examples of controllers can include: (i) microcontrollers, including small, low-power controllers that are commonly used in embedded systems and Internet of Things (IoT) devices; (ii) programmable logic controllers (PLCs) that may be configured to be used in industrial automation systems to control and monitor

manufacturing processes; (iii) system-on-a-chip (SoC) controllers that integrate multiple components such as processors, memory, I/O interfaces, and other peripherals into a single chip; and/or (iv) DSPs.

[0113] A power system of an electronic device may be configured to convert incoming electrical power into a form that can be used to operate the device. A power system can include various components, such as (i) a power source, which can be an alternating current (AC) adapter or a direct current (DC) adapter power supply, (ii) a charger input, which can be configured to use a wired and/or wireless connection (which may be part of a peripheral interface, such as a USB, micro-USB interface, near-field magnetic coupling, magnetic inductive and magnetic resonance charging, and/or radio frequency (RF) charging), (iii) a power-management integrated circuit, configured to distribute power to various components of the device and to ensure that the device operates within safe limits (e.g., regulating voltage, controlling current flow, and/or managing heat dissipation), and/or (iv) a battery configured to store power to provide usable power to components of one or more electronic devices.

[0114] Peripheral interfaces may be electronic components (e.g., of electronic devices) that allow electronic devices to communicate with other devices or peripherals and can provide the ability to input and output data and signals. Examples of peripheral interfaces can include (i) universal serial bus (USB) and/or micro-USB interfaces configured for connecting devices to an electronic device, (ii) Bluetooth interfaces configured to allow devices to communicate with each other, including Bluetooth low energy (BLE), (iii) near field communication (NFC) interfaces configured to be short-range wireless interfaces for operations such as access control, (iv) POGO pins, which may be small, spring-loaded pins configured to provide a charging interface, (v) wireless charging interfaces, (vi) GPS interfaces, (vii) Wi-Fi interfaces for providing a connection between a device and a wireless network, and/or (viii) sensor interfaces.

[0115] Sensors may be electronic components (e.g., in and/or otherwise in electronic communication with electronic devices, such as wearable devices) configured to detect physical and environmental changes and generate electrical signals. Examples of sensors can include (i) imaging sensors for collecting imaging data (e.g., including one or more cameras disposed on a respective electronic device), (ii) biopotential-signal sensors, (iii) inertial measurement units (e.g., IMUs) for detecting, for example, angular rate, force, magnetic field, and/or changes in acceleration, (iv) heart rate sensors for measuring a user's heart rate, (v) SpO₂ sensors for measuring blood oxygen saturation and/or other biometric data of a user, (vi) capacitive sensors for detecting changes in potential at a portion of a user's body (e.g., a sensor-skin interface), and/or (vii) light sensors (e.g., time-of-flight sensors, infrared light sensors, visible light sensors, etc.).

[0116] Biopotential-signal-sensing components may be devices used to measure electrical activity within the body (e.g., biopotential-signal sensors). Some types of biopotential-signal sensors include (i) electroencephalography (EEG) sensors configured to measure electrical activity in the brain to diagnose neurological disorders, (ii) electrocardiogram sensors configured to measure electrical activity of the heart to diagnose heart problems, (iii) electromyography (EMG)

sensors configured to measure the electrical activity of muscles and to diagnose neuromuscular disorders, and (iv) electrooculography (EOG) sensors configured to measure the electrical activity of eye muscles to detect eye movement and diagnose eye disorders.

[0117] An application stored in memory of an electronic device (e.g., software) may include instructions stored in the memory. Examples of such applications include (i) games, (ii) word processors, (iii) messaging applications, (iv) media-streaming applications, (v) financial applications, (vi) calendars, (vii) clocks, and (viii) communication interface modules for enabling wired and/or wireless connections between different respective electronic devices (e.g., IEEE 1502.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, or MiWi), custom or standard wired protocols (e.g., Ethernet or HomePlug), and/or any other suitable communication protocols).

[0118] A communication interface may be a mechanism that enables different systems or devices to exchange information and data with each other, including hardware, software, or a combination of both hardware and software. For example, a communication interface can refer to a physical connector and/or port on a device that enables communication with other devices (e.g., USB, Ethernet, HDMI, Bluetooth). In some embodiments, a communication interface can refer to a software layer that enables different software programs to communicate with each other (e.g., application programming interfaces (APIs), protocols like HTTP and TCP/IP, etc.).

[0119] A graphics module may be a component or software module that is designed to handle graphical operations and/or processes and can include a hardware module and/or a software module.

[0120] Non-transitory computer-readable storage media may be physical devices or storage media that can be used to store electronic data in a non-transitory form (e.g., such that the data is stored permanently until it is intentionally deleted or modified).

[0121] FIGS. 13 and 14 illustrate an example wrist-wearable device 1300 and an example computer system 1400, in accordance with some embodiments. Wrist-wearable device 1300 is an instance of wearable device 902 described in FIG. 9 herein, such that the wearable device 902 should be understood to have the features of the wrist-wearable device 1300 and vice versa. FIG. 14 illustrates components of the wrist-wearable device 1300, which can be used individually or in combination, including combinations that include other electronic devices and/or electronic components.

[0122] FIG. 13 shows a wearable band 1310 and a watch body 1320 (or capsule) being coupled, as discussed below, to form wrist-wearable device 1300. Wrist-wearable device 1300 can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications as well as the functions and/or operations described above with reference to FIGS. 9-12B.

[0123] As will be described in more detail below, operations executed by wrist-wearable device 1300 can include (i) presenting content to a user (e.g., displaying visual content via a display 1305), (ii) detecting (e.g., sensing) user input (e.g., sensing a touch on peripheral button 1323 and/or at a touch screen of the display 1305, a hand gesture detected by sensors (e.g., biopotential sensors)), (iii) sensing biometric data (e.g., neuromuscular signals, heart rate, temperature, sleep, etc.) via one or more sensors 1313, messaging (e.g.,

text, speech, video, etc.); image capture via one or more imaging devices or cameras **1325**, wireless communications (e.g., cellular, near field, Wi-Fi, personal area network, etc.), location determination, financial transactions, providing haptic feedback, providing alarms, providing notifications, providing biometric authentication, providing health monitoring, providing sleep monitoring, etc.

[0124] The above-example functions can be executed independently in watch body **1320**, independently in wearable band **1310**, and/or via an electronic communication between watch body **1320** and wearable band **1310**. In some embodiments, functions can be executed on wrist-wearable device **1300** while an AR environment is being presented (e.g., via one of AR systems **900** to **1200**). The wearable devices described herein can also be used with other types of AR environments.

[0125] Wearable band **1310** can be configured to be worn by a user such that an inner surface of a wearable structure **1311** of wearable band **1310** is in contact with the user's skin. In this example, when worn by a user, sensors **1313** may contact the user's skin. In some examples, one or more of sensors **1313** can sense biometric data such as a user's heart rate, a saturated oxygen level, temperature, sweat level, neuromuscular signals, or a combination thereof. One or more of sensors **1313** can also sense data about a user's environment including a user's motion, altitude, location, orientation, gait, acceleration, position, or a combination thereof. In some embodiment, one or more of sensors **1313** can be configured to track a position and/or motion of wearable band **1310**. One or more of sensors **1313** can include any of the sensors defined above and/or discussed below with respect to FIG. 13.

[0126] One or more of sensors **1313** can be distributed on an inside and/or an outside surface of wearable band **1310**. In some embodiments, one or more of sensors **1313** are uniformly spaced along wearable band **1310**. Alternatively, in some embodiments, one or more of sensors **1313** are positioned at distinct points along wearable band **1310**. As shown in FIG. 13, one or more of sensors **1313** can be the same or distinct. For example, in some embodiments, one or more of sensors **1313** can be shaped as a pill (e.g., sensor **1313a**), an oval, a circle a square, an oblong (e.g., sensor **1313c**) and/or any other shape that maintains contact with the user's skin (e.g., such that neuromuscular signal and/or other biometric data can be accurately measured at the user's skin). In some embodiments, one or more sensors of **1313** are aligned to form pairs of sensors (e.g., for sensing neuromuscular signals based on differential sensing within each respective sensor). For example, sensor **1313b** may be aligned with an adjacent sensor to form sensor pair **1314a** and sensor **1313d** may be aligned with an adjacent sensor to form sensor pair **1314b**. In some embodiments, wearable band **1310** does not have a sensor pair. Alternatively, in some embodiments, wearable band **1310** has a predetermined number of sensor pairs (one pair of sensors, three pairs of sensors, four pairs of sensors, six pairs of sensors, sixteen pairs of sensors, etc.).

[0127] Wearable band **1310** can include any suitable number of sensors **1313**. In some embodiments, the number and arrangement of sensors **1313** depends on the particular application for which wearable band **1310** is used. For instance, wearable band **1310** can be configured as an armband, wristband, or chest-band that include a plurality of sensors **1313** with different number of sensors **1313**, a

variety of types of individual sensors with the plurality of sensors **1313**, and different arrangements for each use case, such as medical use cases as compared to gaming or general day-to-day use cases.

[0128] In accordance with some embodiments, wearable band **1310** further includes an electrical ground electrode and a shielding electrode. The electrical ground and shielding electrodes, like the sensors **1313**, can be distributed on the inside surface of the wearable band **1310** such that they contact a portion of the user's skin. For example, the electrical ground and shielding electrodes can be at an inside surface of a coupling mechanism **1316** or an inside surface of a wearable structure **1311**. The electrical ground and shielding electrodes can be formed and/or use the same components as sensors **1313**. In some embodiments, wearable band **1310** includes more than one electrical ground electrode and more than one shielding electrode.

[0129] Sensors **1313** can be formed as part of wearable structure **1311** of wearable band **1310**. In some embodiments, sensors **1313** are flush or substantially flush with wearable structure **1311** such that they do not extend beyond the surface of wearable structure **1311**. While flush with wearable structure **1311**, sensors **1313** are still configured to contact the user's skin (e.g., via a skin-contacting surface). Alternatively, in some embodiments, sensors **1313** extend beyond wearable structure **1311** a predetermined distance (e.g., 0.1-2 mm) to make contact and depress into the user's skin. In some embodiment, sensors **1313** are coupled to an actuator (not shown) configured to adjust an extension height (e.g., a distance from the surface of wearable structure **1311**) of sensors **1313** such that sensors **1313** make contact and depress into the user's skin. In some embodiments, the actuators adjust the extension height between 0.01 mm-1.2 mm. This may allow a the user to customize the positioning of sensors **1313** to improve the overall comfort of the wearable band **1310** when worn while still allowing sensors **1313** to contact the user's skin. In some embodiments, sensors **1313** are indistinguishable from wearable structure **1311** when worn by the user.

[0130] Wearable structure **1311** can be formed of an elastic material, elastomers, etc., configured to be stretched and fitted to be worn by the user. In some embodiments, wearable structure **1311** is a textile or woven fabric. As described above, sensors **1313** can be formed as part of a wearable structure **1311**. For example, sensors **1313** can be molded into the wearable structure **1311**, be integrated into a woven fabric (e.g., sensors **1313** can be sewn into the fabric and mimic the pliability of fabric and can and/or be constructed from a series woven strands of fabric).

[0131] Wearable structure **1311** can include flexible electronic connectors that interconnect sensors **1313**, the electronic circuitry, and/or other electronic components (described below in reference to FIG. 14) that are enclosed in wearable band **1310**. In some embodiments, the flexible electronic connectors are configured to interconnect sensors **1313**, the electronic circuitry, and/or other electronic components of wearable band **1310** with respective sensors and/or other electronic components of another electronic device (e.g., watch body **1320**). The flexible electronic connectors are configured to move with wearable structure **1311** such that the user adjustment to wearable structure **1311** (e.g., resizing, pulling, folding, etc.) does not stress or strain the electrical coupling of components of wearable band **1310**.

[0132] As described above, wearable band 1310 is configured to be worn by a user. In particular, wearable band 1310 can be shaped or otherwise manipulated to be worn by a user. For example, wearable band 1310 can be shaped to have a substantially circular shape such that it can be configured to be worn on the user's lower arm or wrist. Alternatively, wearable band 1310 can be shaped to be worn on another body part of the user, such as the user's upper arm (e.g., around a bicep), forearm, chest, legs, etc. Wearable band 1310 can include a retaining mechanism 1312 (e.g., a buckle, a hook and loop fastener, etc.) for securing wearable band 1310 to the user's wrist or other body part. While wearable band 1310 is worn by the user, sensors 1313 sense data (referred to as sensor data) from the user's skin. In some examples, sensors 1313 of wearable band 1310 obtain (e.g., sense and record) neuromuscular signals.

[0133] The sensed data (e.g., sensed neuromuscular signals) can be used to detect and/or determine the user's intention to perform certain motor actions. In some examples, sensors 1313 may sense and record neuromuscular signals from the user as the user performs muscular activations (e.g., movements, gestures, etc.). The detected and/or determined motor actions (e.g., phalange (or digit) movements, wrist movements, hand movements, and/or other muscle intentions) can be used to determine control commands or control information (instructions to perform certain commands after the data is sensed) for causing a computing device to perform one or more input commands. For example, the sensed neuromuscular signals can be used to control certain user interfaces displayed on display 1305 of wrist-wearable device 1300 and/or can be transmitted to a device responsible for rendering an artificial-reality environment (e.g., a head-mounted display) to perform an action in an associated artificial-reality environment, such as to control the motion of a virtual device displayed to the user. The muscular activations performed by the user can include static gestures, such as placing the user's hand palm down on a table, dynamic gestures, such as grasping a physical or virtual object, and covert gestures that are imperceptible to another person, such as slightly tensing a joint by co-contracting opposing muscles or using sub-muscular activations. The muscular activations performed by the user can include symbolic gestures (e.g., gestures mapped to other gestures, interactions, or commands, for example, based on a gesture vocabulary that specifies the mapping of gestures to commands).

[0134] The sensor data sensed by sensors 1313 can be used to provide a user with an enhanced interaction with a physical object (e.g., devices communicatively coupled with wearable band 1310) and/or a virtual object in an artificial-reality application generated by an artificial-reality system (e.g., user interface objects presented on the display 1305, or another computing device (e.g., a smartphone)).

[0135] In some embodiments, wearable band 1310 includes one or more haptic devices 1446 (e.g., a vibratory haptic actuator) that are configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user's skin. Sensors 1313 and/or haptic devices 1446 (shown in FIG. 14) can be configured to operate in conjunction with multiple applications including, without limitation, health monitoring, social media, games, and artificial reality (e.g., the applications associated with artificial reality).

[0136] Wearable band 1310 can also include coupling mechanism 1316 for detachably coupling a capsule (e.g., a

computing unit) or watch body 1320 (via a coupling surface of the watch body 1320) to wearable band 1310. For example, a cradle or a shape of coupling mechanism 1316 can correspond to shape of watch body 1320 of wrist-wearable device 1300. In particular, coupling mechanism 1316 can be configured to receive a coupling surface proximate to the bottom side of watch body 1320 (e.g., a side opposite to a front side of watch body 1320 where display 1305 is located), such that a user can push watch body 1320 downward into coupling mechanism 1316 to attach watch body 1320 to coupling mechanism 1316. In some embodiments, coupling mechanism 1316 can be configured to receive a top side of the watch body 1320 (e.g., a side proximate to the front side of watch body 1320 where display 1305 is located) that is pushed upward into the cradle, as opposed to being pushed downward into coupling mechanism 1316. In some embodiments, coupling mechanism 1316 is an integrated component of wearable band 1310 such that wearable band 1310 and coupling mechanism 1316 are a single unitary structure. In some embodiments, coupling mechanism 1316 is a type of frame or shell that allows watch body 1320 coupling surface to be retained within or on wearable band 1310 coupling mechanism 1316 (e.g., a cradle, a tracker band, a support base, a clasp, etc.).

[0137] Coupling mechanism 1316 can allow for watch body 1320 to be detachably coupled to the wearable band 1310 through a friction fit, magnetic coupling, a rotation-based connector, a shear-pin coupler, a retention spring, one or more magnets, a clip, a pin shaft, a hook and loop fastener, or a combination thereof. A user can perform any type of motion to couple the watch body 1320 to wearable band 1310 and to decouple the watch body 1320 from the wearable band 1310. For example, a user can twist, slide, turn, push, pull, or rotate watch body 1320 relative to wearable band 1310, or a combination thereof, to attach watch body 1320 to wearable band 1310 and to detach watch body 1320 from wearable band 1310. Alternatively, as discussed below, in some embodiments, the watch body 1320 can be decoupled from the wearable band 1310 by actuation of a release mechanism 1329.

[0138] Wearable band 1310 can be coupled with watch body 1320 to increase the functionality of wearable band 1310 (e.g., converting wearable band 1310 into wrist-wearable device 1300, adding an additional computing unit and/or battery to increase computational resources and/or a battery life of wearable band 1310, adding additional sensors to improve sensed data, etc.). As described above, wearable band 1310 and coupling mechanism 1316 are configured to operate independently (e.g., execute functions independently) from watch body 1320. For example, coupling mechanism 1316 can include one or more sensors 1313 that contact a user's skin when wearable band 1310 is worn by the user, with or without watch body 1320 and can provide sensor data for determining control commands.

[0139] A user can detach watch body 1320 from wearable band 1310 to reduce the encumbrance of wrist-wearable device 1300 to the user. For embodiments in which watch body 1320 is removable, watch body 1320 can be referred to as a removable structure, such that in these embodiments wrist-wearable device 1300 includes a wearable portion (e.g., wearable band 1310) and a removable structure (e.g., watch body 1320).

[0140] Turning to watch body 1320, in some examples watch body 1320 can have a substantially rectangular or

circular shape. Watch body 1320 is configured to be worn by the user on their wrist or on another body part. More specifically, watch body 1320 is sized to be easily carried by the user, attached on a portion of the user's clothing, and/or coupled to wearable band 1310 (forming the wrist-wearable device 1300). As described above, watch body 1320 can have a shape corresponding to coupling mechanism 1316 of wearable band 1310. In some embodiments, watch body 1320 includes a single release mechanism 1329 or multiple release mechanisms (e.g., two release mechanisms 1329 positioned on opposing sides of watch body 1320, such as spring-loaded buttons) for decoupling watch body 1320 from wearable band 1310. Release mechanism 1329 can include, without limitation, a button, a knob, a plunger, a handle, a lever, a fastener, a clasp, a dial, a latch, or a combination thereof.

[0141] A user can actuate release mechanism 1329 by pushing, turning, lifting, depressing, shifting, or performing other actions on release mechanism 1329. Actuation of release mechanism 1329 can release (e.g., decouple) watch body 1320 from coupling mechanism 1316 of wearable band 1310, allowing the user to use watch body 1320 independently from wearable band 1310 and vice versa. For example, decoupling watch body 1320 from wearable band 1310 can allow a user to capture images using rear-facing camera 1325b. Although release mechanism 1329 is shown positioned at a corner of watch body 1320, release mechanism 1329 can be positioned anywhere on watch body 1320 that is convenient for the user to actuate. In addition, in some embodiments, wearable band 1310 can also include a respective release mechanism for decoupling watch body 1320 from coupling mechanism 1316. In some embodiments, release mechanism 1329 is optional and watch body 1320 can be decoupled from coupling mechanism 1316 as described above (e.g., via twisting, rotating, etc.).

[0142] Watch body 1320 can include one or more peripheral buttons 1323 and 1327 for performing various operations at watch body 1320. For example, peripheral buttons 1323 and 1327 can be used to turn on or wake (e.g., transition from a sleep state to an active state) display 1305, unlock watch body 1320, increase or decrease a volume, increase or decrease a brightness, interact with one or more applications, interact with one or more user interfaces, etc. Additionally or alternatively, in some embodiments, display 1305 operates as a touch screen and allows the user to provide one or more inputs for interacting with watch body 1320.

[0143] In some embodiments, watch body 1320 includes one or more sensors 1321. Sensors 1321 of watch body 1320 can be the same or distinct from sensors 1313 of wearable band 1310. Sensors 1321 of watch body 1320 can be distributed on an inside and/or an outside surface of watch body 1320. In some embodiments, sensors 1321 are configured to contact a user's skin when watch body 1320 is worn by the user. For example, sensors 1321 can be placed on the bottom side of watch body 1320 and coupling mechanism 1316 can be a cradle with an opening that allows the bottom side of watch body 1320 to directly contact the user's skin. Alternatively, in some embodiments, watch body 1320 does not include sensors that are configured to contact the user's skin (e.g., including sensors internal and/or external to the watch body 1320 that are configured to sense data of watch body 1320 and the surrounding environment). In some

embodiments, sensors 1321 are configured to track a position and/or motion of watch body 1320.

[0144] Watch body 1320 and wearable band 1310 can share data using a wired communication method (e.g., a Universal Asynchronous Receiver/Transmitter (UART), a USB transceiver, etc.) and/or a wireless communication method (e.g., near field communication, Bluetooth, etc.). For example, watch body 1320 and wearable band 1310 can share data sensed by sensors 1313 and 1321, as well as application and device specific information (e.g., active and/or available applications, output devices (e.g., displays, speakers, etc.), input devices (e.g., touch screens, microphones, imaging sensors, etc.).

[0145] In some embodiments, watch body 1320 can include, without limitation, a front-facing camera 1325a and/or a rear-facing camera 1325b, sensors 1321 (e.g., a biometric sensor, an IMU, a heart rate sensor, a saturated oxygen sensor, a neuromuscular signal sensor, an altimeter sensor, a temperature sensor, a bioimpedance sensor, a pedometer sensor, an optical sensor (e.g., imaging sensor 1463), a touch sensor, a sweat sensor, etc.). In some embodiments, watch body 1320 can include one or more haptic devices 1476 (e.g., a vibratory haptic actuator) that is configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user. Sensors 1421 and/or haptic device 1476 can also be configured to operate in conjunction with multiple applications including, without limitation, health monitoring applications, social media applications, game applications, and artificial reality applications (e.g., the applications associated with artificial reality).

[0146] As described above, watch body 1320 and wearable band 1310, when coupled, can form wrist-wearable device 1300. When coupled, watch body 1320 and wearable band 1310 may operate as a single device to execute functions (operations, detections, communications, etc.) described herein. In some embodiments, each device may be provided with particular instructions for performing the one or more operations of wrist-wearable device 1300. For example, in accordance with a determination that watch body 1320 does not include neuromuscular signal sensors, wearable band 1310 can include alternative instructions for performing associated instructions (e.g., providing sensed neuromuscular signal data to watch body 1320 via a different electronic device). Operations of wrist-wearable device 1300 can be performed by watch body 1320 alone or in conjunction with wearable band 1310 (e.g., via respective processors and/or hardware components) and vice versa. In some embodiments, operations of wrist-wearable device 1300, watch body 1320, and/or wearable band 1310 can be performed in conjunction with one or more processors and/or hardware components.

[0147] As described below with reference to the block diagram of FIG. 14, wearable band 1310 and/or watch body 1320 can each include independent resources required to independently execute functions. For example, wearable band 1310 and/or watch body 1320 can each include a power source (e.g., a battery), a memory, data storage, a processor (e.g., a central processing unit (CPU)), communications, a light source, and/or input/output devices.

[0148] FIG. 14 shows block diagrams of a computing system 1430 corresponding to wearable band 1310 and a computing system 1460 corresponding to watch body 1320 according to some embodiments. Computing system 1400 of

wrist-wearable device **1300** may include a combination of components of wearable band computing system **1430** and watch body computing system **1460**, in accordance with some embodiments.

[0149] Watch body **1320** and/or wearable band **1310** can include one or more components shown in watch body computing system **1460**. In some embodiments, a single integrated circuit may include all or a substantial portion of the components of watch body computing system **1460** included in a single integrated circuit. Alternatively, in some embodiments, components of the watch body computing system **1460** may be included in a plurality of integrated circuits that are communicatively coupled. In some embodiments, watch body computing system **1460** may be configured to couple (e.g., via a wired or wireless connection) with wearable band computing system **1430**, which may allow the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0150] Watch body computing system **1460** can include one or more processors **1479**, a controller **1477**, a peripherals interface **1461**, a power system **1495**, and memory (e.g., a memory **1480**).

[0151] Power system **1495** can include a charger input **1496**, a power-management integrated circuit (PMIC) **1497**, and a battery **1498**. In some embodiments, a watch body **1320** and a wearable band **1310** can have respective batteries (e.g., battery **1498** and **1459**) and can share power with each other. Watch body **1320** and wearable band **1310** can receive a charge using a variety of techniques. In some embodiments, watch body **1320** and wearable band **1310** can use a wired charging assembly (e.g., power cords) to receive the charge. Alternatively, or in addition, watch body **1320** and/or wearable band **1310** can be configured for wireless charging. For example, a portable charging device can be designed to mate with a portion of watch body **1320** and/or wearable band **1310** and wirelessly deliver usable power to battery **1498** of watch body **1320** and/or battery **1459** of wearable band **1310**. Watch body **1320** and wearable band **1310** can have independent power systems (e.g., power system **1495** and **1456**, respectively) to enable each to operate independently. Watch body **1320** and wearable band **1310** can also share power (e.g., one can charge the other) via respective PMICs (e.g., PMICs **1497** and **1458**) and charger inputs (e.g., **1457** and **1496**) that can share power over power and ground conductors and/or over wireless charging antennas.

[0152] In some embodiments, peripherals interface **1461** can include one or more sensors **1421**. Sensors **1421** can include one or more coupling sensors **1462** for detecting when watch body **1320** is coupled with another electronic device (e.g., a wearable band **1310**). Sensors **1421** can include one or more imaging sensors **1463** (e.g., one or more of cameras **1425**, and/or separate imaging sensors **1463** (e.g., thermal-imaging sensors)). In some embodiments, sensors **1421** can include one or more SpO₂ sensors **1464**. In some embodiments, sensors **1421** can include one or more biopotential-signal sensors (e.g., EMG sensors **1465**, which may be disposed on an interior, user-facing portion of watch body **1320** and/or wearable band **1310**). In some embodiments, sensors **1421** may include one or more capacitive sensors **1466**. In some embodiments, sensors **1421** may include one or more heart rate sensors **1467**. In some embodiments, sensors **1421** may include one or more IMU sensors **1468**. In some embodiments, one or more IMU

sensors **1468** can be configured to detect movement of a user's hand or other location where watch body **1320** is placed or held.

[0153] In some embodiments, one or more of sensors **1421** may provide an example human-machine interface. For example, a set of neuromuscular sensors, such as EMG sensors **1465**, may be arranged circumferentially around wearable band **1310** with an interior surface of EMG sensors **1465** being configured to contact a user's skin. Any suitable number of neuromuscular sensors may be used (e.g., between 2 and 20 sensors). The number and arrangement of neuromuscular sensors may depend on the particular application for which the wearable device is used. For example, wearable band **1310** can be used to generate control information for controlling an augmented reality system, a robot, controlling a vehicle, scrolling through text, controlling a virtual avatar, or any other suitable control task.

[0154] In some embodiments, neuromuscular sensors may be coupled together using flexible electronics incorporated into the wireless device, and the output of one or more of the sensing components can be optionally processed using hardware signal processing circuitry (e.g., to perform amplification, filtering, and/or rectification). In other embodiments, at least some signal processing of the output of the sensing components can be performed in software such as processors **1479**. Thus, signal processing of signals sampled by the sensors can be performed in hardware, software, or by any suitable combination of hardware and software, as aspects of the technology described herein are not limited in this respect.

[0155] Neuromuscular signals may be processed in a variety of ways. For example, the output of EMG sensors **1465** may be provided to an analog front end, which may be configured to perform analog processing (e.g., amplification, noise reduction, filtering, etc.) on the recorded signals. The processed analog signals may then be provided to an analog-to-digital converter, which may convert the analog signals to digital signals that can be processed by one or more computer processors. Furthermore, although this example is as discussed in the context of interfaces with EMG sensors, the embodiments described herein can also be implemented in wearable interfaces with other types of sensors including, but not limited to, mechanomyography (MMG) sensors, sonomyography (SMG) sensors, and electrical impedance tomography (EIT) sensors.

[0156] In some embodiments, peripherals interface **1461** includes a near-field communication (NFC) component **1469**, a global-position system (GPS) component **1470**, a long-term evolution (LTE) component **1471**, and/or a Wi-Fi and/or Bluetooth communication component **1472**. In some embodiments, peripherals interface **1461** includes one or more buttons **1473** (e.g., peripheral buttons **1323** and **1327** in FIG. 13), which, when selected by a user, cause operation to be performed at watch body **1320**. In some embodiments, the peripherals interface **1461** includes one or more indicators, such as a light emitting diode (LED), to provide a user with visual indicators (e.g., message received, low battery, active microphone and/or camera, etc.).

[0157] Watch body **1320** can include at least one display **1305** for displaying visual representations of information or data to a user, including user-interface elements and/or three-dimensional virtual objects. The display can also include a touch screen for inputting user inputs, such as touch gestures, swipe gestures, and the like. Watch body

1320 can include at least one speaker **1474** and at least one microphone **1475** for providing audio signals to the user and receiving audio input from the user. The user can provide user inputs through microphone **1475** and can also receive audio output from speaker **1474** as part of a haptic event provided by haptic controller **1478**. Watch body **1320** can include at least one camera **1425**, including a front camera **1425a** and a rear camera **1425b**. Cameras **1425** can include ultra-wide-angle cameras, wide angle cameras, fish-eye cameras, spherical cameras, telephoto cameras, depth-sensing cameras, or other types of cameras.

[0158] Watch body computing system **1460** can include one or more haptic controllers **1478** and associated componentry (e.g., haptic devices **1476**) for providing haptic events at watch body **1320** (e.g., a vibrating sensation or audio output in response to an event at the watch body **1320**). Haptic controllers **1478** can communicate with one or more haptic devices **1476**, such as electroacoustic devices, including a speaker of the one or more speakers **1474** and/or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating components (e.g., a component that converts electrical signals into tactile outputs on the device). Haptic controller **1478** can provide haptic events to that are capable of being sensed by a user of watch body **1320**. In some embodiments, one or more haptic controllers **1478** can receive input signals from an application of applications **1482**.

[0159] In some embodiments, wearable band computing system **1430** and/or watch body computing system **1460** can include memory **1480**, which can be controlled by one or more memory controllers of controllers **1477**. In some embodiments, software components stored in memory **1480** include one or more applications **1482** configured to perform operations at the watch body **1320**. In some embodiments, one or more applications **1482** may include games, word processors, messaging applications, calling applications, web browsers, social media applications, media streaming applications, financial applications, calendars, clocks, etc. In some embodiments, software components stored in memory **1480** include one or more communication interface modules **1483** as defined above. In some embodiments, software components stored in memory **1480** include one or more graphics modules **1484** for rendering, encoding, and/or decoding audio and/or visual data and one or more data management modules **1485** for collecting, organizing, and/or providing access to data **1487** stored in memory **1480**. In some embodiments, one or more of applications **1482** and/or one or more modules can work in conjunction with one another to perform various tasks at the watch body **1320**.

[0160] In some embodiments, software components stored in memory **1480** can include one or more operating systems **1481** (e.g., a Linux-based operating system, an Android operating system, etc.). Memory **1480** can also include data **1487**. Data **1487** can include profile data **1488A**, sensor data **1489A**, media content data **1490**, and application data **1491**.

[0161] It should be appreciated that watch body computing system **1460** is an example of a computing system within watch body **1320**, and that watch body **1320** can have more or fewer components than shown in watch body computing system **1460**, can combine two or more components, and/or can have a different configuration and/or arrangement of the components. The various components shown in watch body

computing system **1460** are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application-specific integrated circuits.

[0162] Turning to the wearable band computing system **1430**, one or more components that can be included in wearable band **1310** are shown. Wearable band computing system **1430** can include more or fewer components than shown in watch body computing system **1460**, can combine two or more components, and/or can have a different configuration and/or arrangement of some or all of the components. In some embodiments, all, or a substantial portion of the components of wearable band computing system **1430** are included in a single integrated circuit. Alternatively, in some embodiments, components of wearable band computing system **1430** are included in a plurality of integrated circuits that are communicatively coupled. As described above, in some embodiments, wearable band computing system **1430** is configured to couple (e.g., via a wired or wireless connection) with watch body computing system **1460**, which allows the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0163] Wearable band computing system **1430**, similar to watch body computing system **1460**, can include one or more processors **1449**, one or more controllers **1447** (including one or more haptics controllers **1448**), a peripherals interface **1431** that can include one or more sensors **1413** and other peripheral devices, a power source (e.g., a power system **1456**), and memory (e.g., a memory **1450**) that includes an operating system (e.g., an operating system **1451**), data (e.g., data **1454** including profile data **1488B**, sensor data **1489B**, etc.), and one or more modules (e.g., a communications interface module **1452**, a data management module **1453**, etc.).

[0164] One or more of sensors **1413** can be analogous to sensors **1421** of watch body computing system **1460**. For example, sensors **1413** can include one or more coupling sensors **1432**, one or more SpO2 sensors **1434**, one or more EMG sensors **1435**, one or more capacitive sensors **1436**, one or more heart rate sensors **1437**, and one or more IMU sensors **1438**.

[0165] Peripherals interface **1431** can also include other components analogous to those included in peripherals interface **1461** of watch body computing system **1460**, including an NFC component **1439**, a GPS component **1440**, an LTE component **1441**, a Wi-Fi and/or Bluetooth communication component **1442**, and/or one or more haptic devices **1446** as described above in reference to peripherals interface **1461**. In some embodiments, peripherals interface **1431** includes one or more buttons **1443**, a display **1433**, a speaker **1444**, a microphone **1445**, and a camera **1455**. In some embodiments, peripherals interface **1431** includes one or more indicators, such as an LED.

[0166] It should be appreciated that wearable band computing system **1430** is an example of a computing system within wearable band **1310**, and that wearable band **1310** can have more or fewer components than shown in wearable band computing system **1430**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in wearable band computing system **1430** can be implemented in one or more of a combination of hardware,

software, or firmware, including one or more signal processing and/or application-specific integrated circuits.

[0167] Wrist-wearable device **1300** with respect to FIG. **13** is an example of wearable band **1310** and watch body **1320** coupled together, so wrist-wearable device **1300** will be understood to include the components shown and described for wearable band computing system **1430** and watch body computing system **1460**. In some embodiments, wrist-wearable device **1300** has a split architecture (e.g., a split mechanical architecture, a split electrical architecture, etc.) between watch body **1320** and wearable band **1310**. In other words, all of the components shown in wearable band computing system **1430** and watch body computing system **1460** can be housed or otherwise disposed in a combined wrist-wearable device **1300** or within individual components of watch body **1320**, wearable band **1310**, and/or portions thereof (e.g., a coupling mechanism **1316** of wearable band **1310**).

[0168] The techniques described above can be used with any device for sensing neuromuscular signals but could also be used with other types of wearable devices for sensing neuromuscular signals (such as body-wearable or head-wearable devices that might have neuromuscular sensors closer to the brain or spinal column).

[0169] In some embodiments, wrist-wearable device **1300** can be used in conjunction with a head-wearable device (e.g., AR glasses **1500** and VR system **1610**) and/or an HIPD, and wrist-wearable device **1300** can also be configured to be used to allow a user to control any aspect of the artificial reality (e.g., by using EMG-based gestures to control user interface objects in the artificial reality and/or by allowing a user to interact with the touchscreen on the wrist-wearable device to also control aspects of the artificial reality). Having thus described example wrist-wearable devices, attention will now be turned to example head-wearable devices, such as AR glasses **1500** and VR headset **1610**.

[0170] FIGS. **15** to **17** show example artificial-reality systems, which can be used as or in connection with wrist-wearable device **1300**. In some embodiments, AR system **1500** includes an eyewear device **1502**, as shown in FIG. **15**. In some embodiments, VR system **1610** includes a head-mounted display (HMD) **1612**, as shown in FIGS. **16A** and **16B**. In some embodiments, AR system **1500** and VR system **1610** can include one or more analogous components (e.g., components for presenting interactive artificial-reality environments, such as processors, memory, and/or presentation devices, including one or more displays and/or one or more waveguides), some of which are described in more detail with respect to FIG. **17**. As described herein, a head-wearable device can include components of eyewear device **1502** and/or head-mounted display **1612**. Some embodiments of head-wearable devices do not include any displays, including any of the displays described with respect to AR system **1500** and/or VR system **1610**. While the example artificial-reality systems are respectively described herein as AR system **1500** and VR system **1610**, either or both of the example AR systems described herein can be configured to present fully-immersive virtual-reality scenes presented in substantially all of a user's field of view or subtler augmented-reality scenes that are presented within a portion, less than all, of the user's field of view.

[0171] FIG. **15** show an example visual depiction of AR system **1500**, including an eyewear device **1502** (which may

also be described herein as augmented-reality glasses, and/or smart glasses). AR system **1500** can include additional electronic components that are not shown in FIG. **15**, such as a wearable accessory device and/or an intermediary processing device, in electronic communication or otherwise configured to be used in conjunction with the eyewear device **1502**. In some embodiments, the wearable accessory device and/or the intermediary processing device may be configured to couple with eyewear device **1502** via a coupling mechanism in electronic communication with a coupling sensor **1724** (FIG. **17**), where coupling sensor **1724** can detect when an electronic device becomes physically or electronically coupled with eyewear device **1502**. In some embodiments, eyewear device **1502** can be configured to couple to a housing **1790** (FIG. **17**), which may include one or more additional coupling mechanisms configured to couple with additional accessory devices. The components shown in FIG. **15** can be implemented in hardware, software, firmware, or a combination thereof, including one or more signal-processing components and/or application-specific integrated circuits (ASICs).

[0172] Eyewear device **1502** includes mechanical glasses components, including a frame **1504** configured to hold one or more lenses (e.g., one or both lenses **1506-1** and **1506-2**). One of ordinary skill in the art will appreciate that eyewear device **1502** can include additional mechanical components, such as hinges configured to allow portions of frame **1504** of eyewear device **1502** to be folded and unfolded, a bridge configured to span the gap between lenses **1506-1** and **1506-2** and rest on the user's nose, nose pads configured to rest on the bridge of the nose and provide support for eyewear device **1502**, earpieces configured to rest on the user's ears and provide additional support for eyewear device **1502**, temple arms configured to extend from the hinges to the earpieces of eyewear device **1502**, and the like. One of ordinary skill in the art will further appreciate that some examples of AR system **1500** can include none of the mechanical components described herein. For example, smart contact lenses configured to present artificial reality to users may not include any components of eyewear device **1502**.

[0173] Eyewear device **1502** includes electronic components, many of which will be described in more detail below with respect to FIG. **17**. Some example electronic components are illustrated in FIG. **15**, including acoustic sensors **1525-1**, **1525-2**, **1525-3**, **1525-4**, **1525-5**, and **1525-6**, which can be distributed along a substantial portion of the frame **1504** of eyewear device **1502**. Eyewear device **1502** also includes a left camera **1539A** and a right camera **1539B**, which are located on different sides of the frame **1504**. Eyewear device **1502** also includes a processor **1548** (or any other suitable type or form of integrated circuit) that is embedded into a portion of the frame **1504**.

[0174] FIGS. **16A** and **16B** show a VR system **1610** that includes a head-mounted display (HMD) **1612** (e.g., also referred to herein as an artificial-reality headset, a head-wearable device, a VR headset, etc.), in accordance with some embodiments. As noted, some artificial-reality systems (e.g., AR system **1500**) may, instead of blending an artificial reality with actual reality, substantially replace one or more of a user's visual and/or other sensory perceptions of the real world with a virtual experience (e.g., AR systems **1100** and **1200**).

[0175] HMD 1612 includes a front body 1614 and a frame 1616 (e.g., a strap or band) shaped to fit around a user's head. In some embodiments, front body 1614 and/or frame 1616 include one or more electronic elements for facilitating presentation of and/or interactions with an AR and/or VR system (e.g., displays, IMUs, tracking emitter or detectors). In some embodiments, HMD 1612 includes output audio transducers (e.g., an audio transducer 1618), as shown in FIG. 16B. In some embodiments, one or more components, such as the output audio transducer(s) 1618 and frame 1616, can be configured to attach and detach (e.g., are detachably attachable) to HMD 1612 (e.g., a portion or all of frame 1616, and/or audio transducer 1618), as shown in FIG. 16B. In some embodiments, coupling a detachable component to HMD 1612 causes the detachable component to come into electronic communication with HMD 1612.

[0176] FIGS. 16A and 16B also show that VR system 1610 includes one or more cameras, such as left camera 1639A and right camera 1639B, which can be analogous to left and right cameras 1539A and 1539B on frame 1504 of eyewear device 1502. In some embodiments, VR system 1610 includes one or more additional cameras (e.g., cameras 1639C and 1639D), which can be configured to augment image data obtained by left and right cameras 1639A and 1639B by providing more information. For example, camera 1639C can be used to supply color information that is not discerned by cameras 1639A and 1639B. In some embodiments, one or more of cameras 1639A to 1639D can include an optional IR cut filter configured to remove IR light from being received at the respective camera sensors.

[0177] FIG. 17 illustrates a computing system 1720 and an optional housing 1790, each of which show components that can be included in AR system 1500 and/or VR system 1610. In some embodiments, more or fewer components can be included in optional housing 1790 depending on practical restraints of the respective AR system being described.

[0178] In some embodiments, computing system 1720 can include one or more peripherals interfaces 1722A and/or optional housing 1790 can include one or more peripherals interfaces 1722B. Each of computing system 1720 and optional housing 1790 can also include one or more power systems 1742A and 1742B, one or more controllers 1746 (including one or more haptic controllers 1747), one or more processors 1748A and 1748B (as defined above, including any of the examples provided), and memory 1750A and 1750B, which can all be in electronic communication with each other. For example, the one or more processors 1748A and 1748B can be configured to execute instructions stored in memory 1750A and 1750B, which can cause a controller of one or more of controllers 1746 to cause operations to be performed at one or more peripheral devices connected to peripherals interface 1722A and/or 1722B. In some embodiments, each operation described can be powered by electrical power provided by power system 1742A and/or 1742B.

[0179] In some embodiments, peripherals interface 1722A can include one or more devices configured to be part of computing system 1720, some of which have been defined above and/or described with respect to the wrist-wearable devices shown in FIGS. 13 and 14. For example, peripherals interface 1722A can include one or more sensors 1723A. Some example sensors 1723A include one or more coupling sensors 1724, one or more acoustic sensors 1725, one or more imaging sensors 1726, one or more EMG sensors 1727, one or more capacitive sensors 1728, one or more

IMU sensors 1729, and/or any other types of sensors explained above or described with respect to any other embodiments discussed herein.

[0180] In some embodiments, peripherals interfaces 1722A and 1722B can include one or more additional peripheral devices, including one or more NFC devices 1730, one or more GPS devices 1731, one or more LTE devices 1732, one or more Wi-Fi and/or Bluetooth devices 1733, one or more buttons 1734 (e.g., including buttons that are slidable or otherwise adjustable), one or more displays 1735A and 1735B, one or more speakers 1736A and 1736B, one or more microphones 1737, one or more cameras 1738A and 1738B (e.g., including the left camera 1739A and/or a right camera 1739B), one or more haptic devices 1740, and/or any other types of peripheral devices defined above or described with respect to any other embodiments discussed herein.

[0181] AR systems can include a variety of types of visual feedback mechanisms (e.g., presentation devices). For example, display devices in AR system 1500 and/or VR system 1610 can include one or more liquid-crystal displays (LCDs), light emitting diode (LED) displays, organic LED (OLED) displays, and/or any other suitable types of display screens. Artificial-reality systems can include a single display screen (e.g., configured to be seen by both eyes), and/or can provide separate display screens for each eye, which can allow for additional flexibility for varifocal adjustments and/or for correcting a refractive error associated with a user's vision. Some embodiments of AR systems also include optical subsystems having one or more lenses (e.g., conventional concave or convex lenses, Fresnel lenses, or adjustable liquid lenses) through which a user can view a display screen.

[0182] For example, respective displays 1735A and 1735B can be coupled to each of the lenses 1506-1 and 1506-2 of AR system 1500. Displays 1735A and 1735B may be coupled to each of lenses 1506-1 and 1506-2, which can act together or independently to present an image or series of images to a user. In some embodiments, AR system 1500 includes a single display 1735A or 1735B (e.g., a near-eye display) or more than two displays 1735A and 1735B. In some embodiments, a first set of one or more displays 1735A and 1735B can be used to present an augmented-reality environment, and a second set of one or more display devices 1735A and 1735B can be used to present a virtual-reality environment. In some embodiments, one or more waveguides are used in conjunction with presenting artificial-reality content to the user of AR system 1500 (e.g., as a means of delivering light from one or more displays 1735A and 1735B to the user's eyes). In some embodiments, one or more waveguides are fully or partially integrated into the eyewear device 1502. Additionally, or alternatively to display screens, some artificial-reality systems include one or more projection systems. For example, display devices in AR system 1500 and/or VR system 1610 can include micro-LED projectors that project light (e.g., using a waveguide) into display devices, such as clear combiner lenses that allow ambient light to pass through. The display devices can refract the projected light toward a user's pupil and can enable a user to simultaneously view both artificial-reality content and the real world. Artificial-reality systems can also be configured with any other suitable type or form of image projection system. In some embodiments, one or more

waveguides are provided additionally or alternatively to the one or more display(s) 1735A and 1735B.

[0183] Computing system 1720 and/or optional housing 1790 of AR system 1500 or VR system 1610 can include some or all of the components of a power system 1742A and 1742B. Power systems 1742A and 1742B can include one or more charger inputs 1743, one or more PMICs 1744, and/or one or more batteries 1745A and 1744B.

[0184] Memory 1750A and 1750B may include instructions and data, some or all of which may be stored as non-transitory computer-readable storage media within the memories 1750A and 1750B. For example, memory 1750A and 1750B can include one or more operating systems 1751, one or more applications 1752, one or more communication interface applications 1753A and 1753B, one or more graphics applications 1754A and 1754B, one or more AR processing applications 1755A and 1755B, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0185] Memory 1750A and 1750B also include data 1760A and 1760B, which can be used in conjunction with one or more of the applications discussed above. Data 1760A and 1760B can include profile data 1761, sensor data 1762A and 1762B, media content data 1763A, AR application data 1764A and 1764B, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0186] In some embodiments, controller 1746 of eyewear device 1502 may process information generated by sensors 1723A and/or 1723B on eyewear device 1502 and/or another electronic device within AR system 1500. For example, controller 1746 can process information from acoustic sensors 1525-1 and 1525-2. For each detected sound, controller 1746 can perform a direction of arrival (DOA) estimation to estimate a direction from which the detected sound arrived at eyewear device 1502 of AR system 1500. As one or more of acoustic sensors 1725 (e.g., the acoustic sensors 1525-1, 1525-2) detects sounds, controller 1746 can populate an audio data set with the information (e.g., represented in FIG. 17 as sensor data 1762A and 1762B).

[0187] In some embodiments, a physical electronic connector can convey information between eyewear device 1502 and another electronic device and/or between one or more processors 1548, 1748A, 1748B of AR system 1500 or VR system 1610 and controller 1746. The information can be in the form of optical data, electrical data, wireless data, or any other transmittable data form. Moving the processing of information generated by eyewear device 1502 to an intermediary processing device can reduce weight and heat in the eyewear device, making it more comfortable and safer for a user. In some embodiments, an optional wearable accessory device (e.g., an electronic neckband) is coupled to eyewear device 1502 via one or more connectors. The connectors can be wired or wireless connectors and can include electrical and/or non-electrical (e.g., structural) components. In some embodiments, eyewear device 1502 and the wearable accessory device can operate independently without any wired or wireless connection between them.

[0188] In some situations, pairing external devices, such as an intermediary processing device (e.g., HIPD 906, 1006, 1106) with eyewear device 1502 (e.g., as part of AR system 1500) enables eyewear device 1502 to achieve a similar form factor of a pair of glasses while still providing suffi-

cient battery and computation power for expanded capabilities. Some, or all, of the battery power, computational resources, and/or additional features of AR system 1500 can be provided by a paired device or shared between a paired device and eyewear device 1502, thus reducing the weight, heat profile, and form factor of eyewear device 1502 overall while allowing eyewear device 1502 to retain its desired functionality. For example, the wearable accessory device can allow components that would otherwise be included on eyewear device 1502 to be included in the wearable accessory device and/or intermediary processing device, thereby shifting a weight load from the user's head and neck to one or more other portions of the user's body. In some embodiments, the intermediary processing device has a larger surface area over which to diffuse and disperse heat to the ambient environment. Thus, the intermediary processing device can allow for greater battery and computation capacity than might otherwise have been possible on eyewear device 1502 standing alone. Because weight carried in the wearable accessory device can be less invasive to a user than weight carried in the eyewear device 1502, a user may tolerate wearing a lighter eyewear device and carrying or wearing the paired device for greater lengths of time than the user would tolerate wearing a heavier eyewear device standing alone, thereby enabling an artificial-reality environment to be incorporated more fully into a user's day-to-day activities.

[0189] AR systems can include various types of computer vision components and subsystems. For example, AR system 1500 and/or VR system 1610 can include one or more optical sensors such as two-dimensional (2D) or three-dimensional (3D) cameras, time-of-flight depth sensors, structured light transmitters and detectors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. An AR system can process data from one or more of these sensors to identify a location of a user and/or aspects of the user's real-world physical surroundings, including the locations of real-world objects within the real-world physical surroundings. In some embodiments, the methods described herein are used to map the real world, to provide a user with context about real-world surroundings, and/or to generate digital twins (e.g., interactable virtual objects), among a variety of other functions. For example, FIGS. 16A and 16B show VR system 1610 having cameras 1639A to 1639D, which can be used to provide depth information for creating a voxel field and a two-dimensional mesh to provide object information to the user to avoid collisions.

[0190] In some embodiments, AR system 1500 and/or VR system 1610 can include haptic (tactile) feedback systems, which may be incorporated into headwear, gloves, body suits, handheld controllers, environmental devices (e.g., chairs or floor mats), and/or any other type of device or system, such as the wearable devices discussed herein. The haptic feedback systems may provide various types of cutaneous feedback, including vibration, force, traction, shear, texture, and/or temperature. The haptic feedback systems may also provide various types of kinesthetic feedback, such as motion and compliance. The haptic feedback may be implemented using motors, piezoelectric actuators, fluidic systems, and/or a variety of other types of feedback mechanisms. The haptic feedback systems may be implemented independently of other artificial-reality devices,

within other artificial-reality devices, and/or in conjunction with other artificial-reality devices.

[0191] In some embodiments of an artificial reality system, such as AR system **1500** and/or VR system **1610**, ambient light (e.g., a live feed of the surrounding environment that a user would normally see) can be passed through a display element of a respective head-wearable device presenting aspects of the AR system. In some embodiments, ambient light can be passed through a portion less than all of an AR environment presented within a user's field of view (e.g., a portion of the AR environment co-located with a physical object in the user's real-world environment that is within a designated boundary (e.g., a guardian boundary) configured to be used by the user while they are interacting with the AR environment). For example, a visual user interface element (e.g., a notification user interface element) can be presented at the head-wearable device, and an amount of ambient light (e.g., 15-50% of the ambient light) can be passed through the user interface element such that the user can distinguish at least a portion of the physical environment over which the user interface element is being displayed.

[0192] In some embodiments, the systems described herein may also include an eye-tracking subsystem designed to identify and track various characteristics of a user's eye(s), such as the user's gaze direction. The phrase "eye tracking" may, in some examples, refer to a process by which the position, orientation, and/or motion of an eye is measured, detected, sensed, determined, and/or monitored. The disclosed systems may measure the position, orientation, and/or motion of an eye in a variety of different ways, including through the use of various optical-based eye-tracking techniques, ultrasound-based eye-tracking techniques, etc. An eye-tracking subsystem may be configured in a number of different ways and may include a variety of different eye-tracking hardware components or other computer-vision components. For example, an eye-tracking subsystem may include a variety of different optical sensors, such as two-dimensional (2D) or 3D cameras, time-of-flight depth sensors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. In this example, a processing subsystem may process data from one or more of these sensors to measure, detect, determine, and/or otherwise monitor the position, orientation, and/or motion of the user's eye(s).

[0193] FIG. **18** is an illustration of an example system **1800** that incorporates an eye-tracking subsystem capable of tracking a user's eye(s). As depicted in FIG. **18**, system **1800** may include a light source **1802**, an optical subsystem **1804**, an eye-tracking subsystem **1806**, and/or a control subsystem **1808**. In some examples, light source **1802** may generate light for an image (e.g., to be presented to an eye **1801** of the viewer). Light source **1802** may represent any of a variety of suitable devices. For example, light source **1802** can include a two-dimensional projector (e.g., a LCOS display), a scanning source (e.g., a scanning laser), or other device (e.g., an LCD, an LED display, an OLED display, an active-matrix OLED display (AMOLED), a transparent OLED display (TOLED), a waveguide, or some other display capable of generating light for presenting an image to the viewer). In some examples, the image may represent a virtual image, which may refer to an optical image formed from the

apparent divergence of light rays from a point in space, as opposed to an image formed from the light ray's actual divergence.

[0194] In some embodiments, optical subsystem **1804** may receive the light generated by light source **1802** and generate, based on the received light, converging light **1820** that includes the image. In some examples, optical subsystem **1804** may include any number of lenses (e.g., Fresnel lenses, convex lenses, concave lenses), apertures, filters, mirrors, prisms, and/or other optical components, possibly in combination with actuators and/or other devices. In particular, the actuators and/or other devices may translate and/or rotate one or more of the optical components to alter one or more aspects of converging light **1820**. Further, various mechanical couplings may serve to maintain the relative spacing and/or the orientation of the optical components in any suitable combination.

[0195] In one embodiment, eye-tracking subsystem **1806** may generate tracking information indicating a gaze angle of an eye **1801** of the viewer. In this embodiment, control subsystem **1808** may control aspects of optical subsystem **1804** (e.g., the angle of incidence of converging light **1820**) based at least in part on this tracking information. Additionally, in some examples, control subsystem **1808** may store and utilize historical tracking information (e.g., a history of the tracking information over a given duration, such as the previous second or fraction thereof) to anticipate the gaze angle of eye **1801** (e.g., an angle between the visual axis and the anatomical axis of eye **1801**). In some embodiments, eye-tracking subsystem **1806** may detect radiation emanating from some portion of eye **1801** (e.g., the cornea, the iris, the pupil, or the like) to determine the current gaze angle of eye **1801**. In other examples, eye-tracking subsystem **1806** may employ a wavefront sensor to track the current location of the pupil.

[0196] Any number of techniques can be used to track eye **1801**. Some techniques may involve illuminating eye **1801** with infrared light and measuring reflections with at least one optical sensor that is tuned to be sensitive to the infrared light. Information about how the infrared light is reflected from eye **1801** may be analyzed to determine the position(s), orientation(s), and/or motion(s) of one or more eye feature(s), such as the cornea, pupil, iris, and/or retinal blood vessels.

[0197] In some examples, the radiation captured by a sensor of eye-tracking subsystem **1806** may be digitized (i.e., converted to an electronic signal). Further, the sensor may transmit a digital representation of this electronic signal to one or more processors (for example, processors associated with a device including eye-tracking subsystem **1806**). Eye-tracking subsystem **1806** may include any of a variety of sensors in a variety of different configurations. For example, eye-tracking subsystem **1806** may include an infrared detector that reacts to infrared radiation. The infrared detector may be a thermal detector, a photonic detector, and/or any other suitable type of detector. Thermal detectors may include detectors that react to thermal effects of the incident infrared radiation.

[0198] In some examples, one or more processors may process the digital representation generated by the sensor(s) of eye-tracking subsystem **1806** to track the movement of eye **1801**. In another example, these processors may track the movements of eye **1801** by executing algorithms represented by computer-executable instructions stored on non-

transitory memory. In some examples, on-chip logic (e.g., an application-specific integrated circuit or ASIC) may be used to perform at least portions of such algorithms. As noted, eye-tracking subsystem **1806** may be programmed to use an output of the sensor(s) to track movement of eye **1801**. In some embodiments, eye-tracking subsystem **1806** may analyze the digital representation generated by the sensors to extract eye rotation information from changes in reflections. In one embodiment, eye-tracking subsystem **1806** may use corneal reflections or glints (also known as Purkinje images) and/or the center of the eye's pupil **1822** as features to track over time.

[0199] In some embodiments, eye-tracking subsystem **1806** may use the center of the eye's pupil **1822** and infrared or near-infrared, non-collimated light to create corneal reflections. In these embodiments, eye-tracking subsystem **1806** may use the vector between the center of the eye's pupil **1822** and the corneal reflections to compute the gaze direction of eye **1801**. In some embodiments, the disclosed systems may perform a calibration procedure for an individual (using, e.g., supervised or unsupervised techniques) before tracking the user's eyes. For example, the calibration procedure may include directing users to look at one or more points displayed on a display while the eye-tracking system records the values that correspond to each gaze position associated with each point.

[0200] In some embodiments, eye-tracking subsystem **1806** may use two types of infrared and/or near-infrared (also known as active light) eye-tracking techniques: bright-pupil and dark-pupil eye tracking, which may be differentiated based on the location of an illumination source with respect to the optical elements used. If the illumination is coaxial with the optical path, then eye **1801** may act as a retroreflector as the light reflects off the retina, thereby creating a bright pupil effect similar to a red-eye effect in photography. If the illumination source is offset from the optical path, then the eye's pupil **1822** may appear dark because the retroreflection from the retina is directed away from the sensor. In some embodiments, bright-pupil tracking may create greater iris/pupil contrast, allowing more robust eye tracking with iris pigmentation, and may feature reduced interference (e.g., interference caused by eyelashes and other obscuring features). Bright-pupil tracking may also allow tracking in lighting conditions ranging from total darkness to a very bright environment.

[0201] In some embodiments, control subsystem **1808** may control light source **1802** and/or optical subsystem **1804** to reduce optical aberrations (e.g., chromatic aberrations and/or monochromatic aberrations) of the image that may be caused by or influenced by eye **1801**. In some examples, as mentioned above, control subsystem **1808** may use the tracking information from eye-tracking subsystem **1806** to perform such control. For example, in controlling light source **1802**, control subsystem **1808** may alter the light generated by light source **1802** (e.g., by way of image rendering) to modify (e.g., pre-distort) the image so that the aberration of the image caused by eye **1801** is reduced.

[0202] The disclosed systems may track both the position and relative size of the pupil (since, e.g., the pupil dilates and/or contracts). In some examples, the eye-tracking devices and components (e.g., sensors and/or sources) used for detecting and/or tracking the pupil may be different (or calibrated differently) for different types of eyes. For example, the frequency range of the sensors may be different

(or separately calibrated) for eyes of different colors and/or different pupil types, sizes, and/or the like. As such, the various eye-tracking components (e.g., infrared sources and/or sensors) described herein may need to be calibrated for each individual user and/or eye.

[0203] The disclosed systems may track both eyes with and without ophthalmic correction, such as that provided by contact lenses worn by the user. In some embodiments, ophthalmic correction elements (e.g., adjustable lenses) may be directly incorporated into the artificial reality systems described herein. In some examples, the color of the user's eye may necessitate modification of a corresponding eye-tracking algorithm. For example, eye-tracking algorithms may need to be modified based at least in part on the differing color contrast between a brown eye and, for example, a blue eye.

[0204] FIG. 19 is a more detailed illustration of various aspects of the eye-tracking subsystem illustrated in FIG. 18. As shown in this figure, an eye-tracking subsystem **1900** may include at least one source **1904** and at least one sensor **1906**. Source **1904** generally represents any type or form of element capable of emitting radiation. In one example, source **1904** may generate visible, infrared, and/or near-infrared radiation. In some examples, source **1904** may radiate non-collimated infrared and/or near-infrared portions of the electromagnetic spectrum towards an eye **1902** of a user. Source **1904** may utilize a variety of sampling rates and speeds. For example, the disclosed systems may use sources with higher sampling rates in order to capture fixational eye movements of a user's eye **1902** and/or to correctly measure saccade dynamics of the user's eye **1902**. As noted above, any type or form of eye-tracking technique may be used to track the user's eye **1902**, including optical-based eye-tracking techniques, ultrasound-based eye-tracking techniques, etc.

[0205] Sensor **1906** generally represents any type or form of element capable of detecting radiation, such as radiation reflected off the user's eye **1902**. Examples of sensor **1906** include, without limitation, a charge coupled device (CCD), a photodiode array, a complementary metal-oxide-semiconductor (CMOS) based sensor device, and/or the like. In one example, sensor **1906** may represent a sensor having predetermined parameters, including, but not limited to, a dynamic resolution range, linearity, and/or other characteristic selected and/or designed specifically for eye tracking.

[0206] As detailed above, eye-tracking subsystem **1900** may generate one or more glints. As detailed above, a glint **1903** may represent reflections of radiation (e.g., infrared radiation from an infrared source, such as source **1904**) from the structure of the user's eye. In various embodiments, glint **1903** and/or the user's pupil may be tracked using an eye-tracking algorithm executed by a processor (either within or external to an artificial reality device). For example, an artificial reality device may include a processor and/or a memory device in order to perform eye tracking locally and/or a transceiver to send and receive the data necessary to perform eye tracking on an external device (e.g., a mobile phone, cloud server, or other computing device).

[0207] FIG. 19 shows an example image **1905** captured by an eye-tracking subsystem, such as eye-tracking subsystem **1900**. In this example, image **1905** may include both the user's pupil **1908** and a glint **1910** near the same. In some examples, pupil **1908** and/or glint **1910** may be identified

using an artificial-intelligence-based algorithm, such as a computer-vision-based algorithm. In one embodiment, image **1905** may represent a single frame in a series of frames that may be analyzed continuously in order to track the eye **1902** of the user. Further, pupil **1908** and/or glint **1910** may be tracked over a period of time to determine a user's gaze.

[0208] In one example, eye-tracking subsystem **1900** may be configured to identify and measure the inter-pupillary distance (IPD) of a user. In some embodiments, eye-tracking subsystem **1900** may measure and/or calculate the IPD of the user while the user is wearing the artificial reality system. In these embodiments, eye-tracking subsystem **1900** may detect the positions of a user's eyes and may use this information to calculate the user's IPD.

[0209] As noted, the eye-tracking systems or subsystems disclosed herein may track a user's eye position and/or eye movement in a variety of ways. In one example, one or more light sources and/or optical sensors may capture an image of the user's eyes. The eye-tracking subsystem may then use the captured information to determine the user's inter-pupillary distance, interocular distance, and/or a 3D position of each eye (e.g., for distortion adjustment purposes), including a magnitude of torsion and rotation (i.e., roll, pitch, and yaw) and/or gaze directions for each eye. In one example, infrared light may be emitted by the eye-tracking subsystem and reflected from each eye. The reflected light may be received or detected by an optical sensor and analyzed to extract eye rotation data from changes in the infrared light reflected by each eye.

[0210] The eye-tracking subsystem may use any of a variety of different methods to track the eyes of a user. For example, a light source (e.g., infrared light-emitting diodes) may emit a dot pattern onto each eye of the user. The eye-tracking subsystem may then detect (e.g., via an optical sensor coupled to the artificial reality system) and analyze a reflection of the dot pattern from each eye of the user to identify a location of each pupil of the user. Accordingly, the eye-tracking subsystem may track up to six degrees of freedom of each eye (i.e., 3D position, roll, pitch, and yaw) and at least a subset of the tracked quantities may be combined from two eyes of a user to estimate a gaze point (i.e., a 3D location or position in a virtual scene where the user is looking) and/or an IPD.

[0211] In some cases, the distance between a user's pupil and a display may change as the user's eye moves to look in different directions. The varying distance between a pupil and a display as viewing direction changes may be referred to as "pupil swim" and may contribute to distortion perceived by the user as a result of light focusing in different locations as the distance between the pupil and the display changes. Accordingly, measuring distortion at different eye positions and pupil distances relative to displays and generating distortion corrections for different positions and distances may allow mitigation of distortion caused by pupil swim by tracking the 3D position of a user's eyes and applying a distortion correction corresponding to the 3D position of each of the user's eyes at a given point in time. Thus, knowing the 3D position of each of a user's eyes may allow for the mitigation of distortion caused by changes in the distance between the pupil of the eye and the display by applying a distortion correction for each 3D eye position. Furthermore, as noted above, knowing the position of each

of the user's eyes may also enable the eye-tracking subsystem to make automated adjustments for a user's IPD.

[0212] In some embodiments, a display subsystem may include a variety of additional subsystems that may work in conjunction with the eye-tracking subsystems described herein. For example, a display subsystem may include a varifocal subsystem, a scene-rendering module, and/or a vergence-processing module. The varifocal subsystem may cause left and right display elements to vary the focal distance of the display device. In one embodiment, the varifocal subsystem may physically change the distance between a display and the optics through which it is viewed by moving the display, the optics, or both. Additionally, moving or translating two lenses relative to each other may also be used to change the focal distance of the display. Thus, the varifocal subsystem may include actuators or motors that move displays and/or optics to change the distance between them. This varifocal subsystem may be separate from or integrated into the display subsystem. The varifocal subsystem may also be integrated into or separate from its actuation subsystem and/or the eye-tracking subsystems described herein.

[0213] In one example, the display subsystem may include a vergence-processing module configured to determine a vergence depth of a user's gaze based on a gaze point and/or an estimated intersection of the gaze lines determined by the eye-tracking subsystem. Vergence may refer to the simultaneous movement or rotation of both eyes in opposite directions to maintain single binocular vision, which may be naturally and automatically performed by the human eye. Thus, a location where a user's eyes are verged is where the user is looking and is also typically the location where the user's eyes are focused. For example, the vergence-processing module may triangulate gaze lines to estimate a distance or depth from the user associated with intersection of the gaze lines. The depth associated with intersection of the gaze lines may then be used as an approximation for the accommodation distance, which may identify a distance from the user where the user's eyes are directed. Thus, the vergence distance may allow for the determination of a location where the user's eyes should be focused and a depth from the user's eyes at which the eyes are focused, thereby providing information (such as an object or plane of focus) for rendering adjustments to the virtual scene.

[0214] The vergence-processing module may coordinate with the eye-tracking subsystems described herein to make adjustments to the display subsystem to account for a user's vergence depth. When the user is focused on something at a distance, the user's pupils may be slightly farther apart than when the user is focused on something close. The eye-tracking subsystem may obtain information about the user's vergence or focus depth and may adjust the display subsystem to be closer together when the user's eyes focus or verge on something close and to be farther apart when the user's eyes focus or verge on something at a distance.

[0215] The eye-tracking information generated by the above-described eye-tracking subsystems may also be used, for example, to modify various aspect of how different computer-generated images are presented. For example, a display subsystem may be configured to modify, based on information generated by an eye-tracking subsystem, at least one aspect of how the computer-generated images are presented. For instance, the computer-generated images may be modified based on the user's eye movement, such that if a

user is looking up, the computer-generated images may be moved upward on the screen. Similarly, if the user is looking to the side or down, the computer-generated images may be moved to the side or downward on the screen. If the user's eyes are closed, the computer-generated images may be paused or removed from the display and resumed once the user's eyes are back open.

[0216] The above-described eye-tracking subsystems can be incorporated into one or more of the various artificial reality systems described herein in a variety of ways. For example, one or more of the various components of system 1800 and/or eye-tracking subsystem 1900 may be incorporated into any of the augmented-reality systems in and/or virtual-reality systems described herein in to enable these systems to perform various eye-tracking tasks (including one or more of the eye-tracking operations described herein).

[0217] The process parameters and sequence of the steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein may be shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various exemplary methods described and/or illustrated herein may also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

[0218] The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the exemplary embodiments disclosed herein. This exemplary description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The embodiments disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to the appended claims and their equivalents in determining the scope of the present disclosure.

[0219] Unless otherwise noted, the terms “connected to” and “coupled to” (and their derivatives), as used in the specification and claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms “a” or “an,” as used in the specification and claims, are to be construed as meaning “at least one of.” Finally, for ease of use, the terms “including” and “having” (and their derivatives), as used in the specification and claims, are interchangeable with and have the same meaning as the word “comprising.”

What is claimed is:

1. A device comprising:

a plurality of waveguides consisting of a first plurality of waveguides, a second plurality of waveguides and a third plurality of waveguides; and

an optical dispatching circuit comprising a first input, a second input and a third input, the optical dispatching circuit configured to direct light from the first input through each of the first plurality of waveguides, to direct light from the second input through each of the second plurality of waveguides, and to direct light from the third input through each of the third plurality of waveguides.

2. The device of claim 1, wherein the optical dispatching circuit comprises:

an input multiplexer coupled to the first input, the second input and the third input and comprising an output;

a bus waveguide coupled to the output of the input multiplexer; and

a transport slab arranged adjacent to and apart from the bus waveguide, the transport slab coupled to each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

3. The device of claim 2, wherein a distance between the bus waveguide and the transport slab narrows from a first end of the bus waveguide proximate to the input multiplexer to an opposing end of the bus waveguide distal from the input multiplexer.

4. The device of claim 2, further comprising a plurality of output multiplexers that couple the transport slab to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

5. The device of claim 2, wherein the input multiplexer comprises:

a first multiplexer waveguide coupled to the first input;

a second multiplexer waveguide coupled to the second input; and

a third multiplexer waveguide coupled to the third input and coupled to the output of the input multiplexer,

wherein the first multiplexer waveguide widens from the first input to the output of the input multiplexer,

wherein the second multiplexer waveguide narrows from the second input to the output of the input multiplexer, and

wherein the third multiplexer waveguide narrows from the third input to the output of the input multiplexer.

6. The device of claim 1, wherein the optical dispatching circuit comprises a waveguide tree coupled to the first input, the second input and the third input, and comprising a plurality of outputs that includes an output for each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

7. The device of claim 6, wherein the waveguide tree comprises a plurality of Y-shaped waveguide sections.

8. The device of claim 6, wherein the waveguide tree comprises a plurality of directional couplers, each directional coupler configured to swap spatial modes of light across a pair of waveguides therein.

9. The device of claim 8, wherein, in each directional coupler of the plurality of directional couplers, one of the pair of waveguides tapers narrower across the directional coupler from an input side to an output side while another of the pair of waveguides tapers wider from the input side to the output side of the directional coupler.

10. The device of claim 1, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

11. The device of claim 10, further comprising a plurality of output multiplexers that couple each of the plurality of receiving waveguides to the first plurality of waveguides, the second plurality of waveguides and the third plurality of

waveguides, wherein each receiving waveguide of the plurality of receiving waveguides is coupled to a respective output multiplexer of the plurality of output multiplexers, and wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

12. The device of claim **10**, wherein a width of each receiving waveguide of the plurality of receiving waveguides increases with distance from a central axis of the coupler waveguide.

13. The device of claim **10**, further comprising:

a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide;

a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide;

a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide,

wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

14. The device of claim **1**, further comprising:

a first light source arranged to direct light of a first wavelength through the first input of the optical dispatching circuit;

a second light source arranged to direct light of a second wavelength, different to the first wavelength, through the second input of the optical dispatching circuit; and

a third light source arranged to direct light of a third wavelength, different to the first wavelength and the second wavelength, through the third input of the optical dispatching circuit.

15. The device of claim **14**, wherein the first wavelength is between 600 nm and 700 nm, the second wavelength is between 500 nm and 600 nm, and the third wavelength is between 400 nm and 500 nm.

16. The device of claim **1**, wherein each of the plurality of waveguides comprises a plurality of emitter structures configured to output light.

17. A display comprising the device of claim **16** and a liquid crystal on silicon (LCoS) panel arranged to receive light output from the plurality of emitter structures.

18. A method comprising:

directing light from a first light source into a first input of an optical dispatching circuit, through the optical dispatching circuit, into a first plurality of waveguides and into a first plurality of emitter structures arranged within the first plurality of waveguides;

directing light from a second light source into a second input of the optical dispatching circuit, through the optical dispatching circuit, into a second plurality of waveguides and into a second plurality of emitter structures arranged within the second plurality of waveguides; and

directing light from a third light source into a third input of the optical dispatching circuit, through the optical dispatching circuit, into a third plurality of waveguides and into a third plurality of emitter structures arranged within the third plurality of waveguides.

19. The method of claim **18**, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

20. The method of claim **19**, wherein:

directing light from the first light source into the first input of the optical dispatching circuit comprises directing the light from the first light source through a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide;

directing light from the second light source into the second input of the optical dispatching circuit comprises directing the light from the second light source through a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; and

directing light from the third light source into the third input of the optical dispatching circuit comprises directing the light from the third light source through a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide.

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